Nkumba University

CHARTERED SINCE 2006 BY THE GOVERNMENT OF UGANDA

SCHOOL OF COMPUTING AND INFORMATICS TECHNOLOGY

The mission of the University is to provide an environment that enables the cultivation of Competence, Confidence, Creativity, and Character, in the academic, professional and social interactions.

FINAL DISSERTATION

DESIGN AND IMPLEMENTATION OF A REAL-TIME PARKING MANAGEMENT AND SPACE RESERVATION SYSTEM: A CASE STUDY OF KAMPALA

BY

NANYONJO JACKIE

-----------WKD

STUDENT NUMBER

JULY, 2026

PRESENTED TO

THE SCHOOL OF COMPUTING AND INFORMATICS

IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF A BACHELOR’S DEGREE IN INFORMATION SYSTEMS & TECHNOLOGY (BIST) OF NKUMBA UNIVERSITY.

DECLARATION

I, **Nanyonjo Jackie**, hereby declare that this dissertation titled **"Design and Implementation of a Real-Time Parking Management and Space Reservation System: A Case Study of Kampala"** is my original work and has not been submitted to any institution of higher learning for any academic award.

All sources of information used in this study have been acknowledged and properly cited in accordance with academic requirements. The work presented in this dissertation was carried out under the guidance and supervision of my academic supervisor.

Student's Name: Nanyonjo Jackie

Registration/Index Number:

Signature: ________________________________

Date: ________________________________

APPROVAL

This dissertation titled **"Design and Implementation of a Real-Time Parking Management and Space Reservation System: A Case Study of Kampala"** has been prepared and submitted by **Nanyonjo Jackie** under my supervision. It is hereby approved for submission to the **School of Computing and Informatics Technology, Nkumba University**, in partial fulfillment of the requirements for the award of a **Bachelor's Degree in Information Systems and Technology**.

Supervisor's Name: Ms. Nassozi Betty

Signature: ________________________________

Date: ________________________________

DEDICATION

I dedicate this dissertation to my family, whose love, encouragement, patience, and support have been a constant source of strength throughout my academic journey. Their belief in my ability to succeed has motivated me to remain focused and committed, even during challenging moments.

I also dedicate this work to my friends, classmates, and all those who supported me in different ways during the course of this study. Their encouragement, ideas, and cooperation contributed greatly to the successful completion of this research and system development project.

Above all, I dedicate this work to God Almighty, whose grace, wisdom, protection, and guidance enabled me to complete this dissertation successfully.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to God Almighty for the gift of life, wisdom, strength, and guidance throughout the course of this study. His grace enabled me to complete this dissertation and the development of the Parka system.

I extend my deepest appreciation to my academic supervisor, **Ms. Nassozi Betty**, for the guidance, corrections, encouragement, and professional advice provided throughout this research process. The supervisor's support helped me improve the quality of this dissertation and remain focused on the objectives of the study.

I am also grateful to the management, lecturers, and staff of the **School of Computing and Informatics Technology, Nkumba University**, for the knowledge, skills, and academic support offered during my course of study. The training received throughout the course provided the foundation for carrying out this project.

Special thanks go to my family for their love, patience, prayers, and continuous support. Their encouragement gave me strength during the demanding stages of research, system development, and report writing.

I also appreciate my classmates, friends, and colleagues who shared ideas, gave feedback, and supported me in different ways during the development and completion of this project.

Finally, I acknowledge all respondents and individuals who contributed information, feedback, and insights that helped in understanding the parking management challenges in Kampala and in shaping the design and implementation of Parka.

Table of Contents

ABSTRACT

This study focused on the **design and implementation of a real-time parking management and space reservation system: A case study of Kampala**. The study was motivated by the increasing parking challenges experienced in Kampala, where drivers often spend time searching for available parking spaces due to limited real-time information, absence of advance reservation mechanisms, manual parking operations, and poor digital record keeping. These challenges contribute to time wastage, fuel consumption, traffic congestion, and inefficiency in parking facility management.

The main objective of the study was to design and implement a real-time parking management and space reservation system to improve parking efficiency in Kampala. The specific objectives were to review the current parking management system and identify gaps, design a system that addresses the identified gaps, implement the designed system using suitable web technologies, and evaluate the effectiveness of the developed system.

The study adopted a case study research design supported by descriptive and developmental approaches. Data was collected using questionnaires, interviews, observation, and secondary sources. The collected data helped in identifying the weaknesses of the existing parking system and defining the requirements of the proposed system. Agile methodology was used during system development because it allowed iterative development, continuous testing, and improvement of system features.

The developed system, named **Parka**, was implemented as a web-based real-time parking management and reservation platform. The frontend was developed using React, Vite, TypeScript, and Tailwind CSS, while the backend was developed using Node.js and Express. PostgreSQL was used as the database, and Socket.io was used to support real-time updates and notifications. The system supports four main user roles: drivers, parking attendants, parking facility operators, and system administrators. Key features include user authentication, parking facility search, real-time availability display, parking reservation, QR-based check-in and check-out, vehicle management, fee calculation, payment simulation, receipt generation, notifications, facility management, and administrator monitoring.

The system was evaluated in terms of functionality, usability, performance, security, reliability, and data integrity. The evaluation showed that Parka successfully supported parking search, reservation, QR-based verification, session tracking, payment simulation, receipt generation, and role-based parking management. The study concluded that a web-based real-time parking management and space reservation system can improve parking efficiency by reducing uncertainty for drivers, improving space utilization, supporting better operator visibility, and enhancing accountability in parking operations.

Keywords: Parking management, real-time availability, parking reservation, smart parking, Parka, Kampala, web-based system, QR check-in, PostgreSQL, React.

CHAPTER ONE: INTRODUCTION

1.1 Introduction

This chapter presents the background of the study, statement of the problem, objectives of the study, research questions, scope of the study, significance of the study, limitations, and the conceptual framework. The chapter introduces the parking management challenges experienced in Kampala and explains the need for a real-time parking management and space reservation system.

The study focused on the design and implementation of **Parka**, a web-based real-time parking management and space reservation system. The system was developed to help drivers search for parking facilities, view parking availability, reserve spaces, check in and check out digitally, make simulated payments, and receive receipts. It was also designed to help parking attendants, operators, and administrators manage parking operations more efficiently.

1.2 Background of the Study

Urbanization has led to an increase in the number of vehicles in cities across the world. As urban populations grow and economic activities expand, more people use private vehicles, taxis, delivery vehicles, and motorcycles to move within cities. This growth increases the demand for parking spaces, especially in commercial and administrative areas where many people travel daily.

Parking management has therefore become an important part of urban transport planning. Poor parking management contributes to traffic congestion, fuel wastage, environmental pollution, delays, and poor user experience. When drivers spend time moving around in search of parking spaces, they increase traffic pressure on already busy roads. Shoup (2006) noted that vehicles cruising for parking contribute to congestion because drivers remain in circulation while looking for available spaces.

In many developing cities, parking management is still largely manual. Drivers often depend on physical observation, roadside attendants, and trial-and-error movement from one parking area to another. This creates uncertainty because a driver may only know whether a parking space is available after reaching the destination. If the facility is full, the driver must continue searching elsewhere.

Kampala, the capital city of Uganda, faces similar parking management challenges. The city has experienced increased vehicle movement due to commercial growth, population growth, and concentration of services in the central business district and surrounding urban areas. The proposal for this study identified that Kampala's parking systems are largely manual and lack real-time information sharing, making it difficult for drivers to know where parking spaces are available before arrival.

The existing parking system in Kampala depends heavily on parking attendants and cash-based payments. Although attendants support parking operations, manual processes make it difficult to provide real-time availability, advance reservation, digital records, automated fee calculation, and centralized monitoring. Parking operators may also find it difficult to track occupancy, reservations, payments, attendants, and revenue accurately.

Information and communication technologies have created opportunities for improving parking management. Smart parking systems use digital platforms, real-time data, mobile applications, web systems, sensors, QR codes, digital payments, and dashboards to improve parking availability, reservation, monitoring, and reporting. Litman (2025) explains that parking management strategies can improve the efficient use of parking resources and reduce unnecessary parking-related costs.

Parka was developed in response to these challenges. The system was designed as a real-time parking management and space reservation platform for Kampala. It provides drivers with digital access to parking information and allows them to reserve spaces before arrival. It also supports parking attendants in verifying reservations and managing check-in and check-out, while operators and administrators can monitor and control parking activities through dashboards.

1.3 Statement of the Problem

Parking in Kampala has become increasingly inefficient due to the growing number of vehicles, limited organized parking infrastructure, and continued dependence on manual parking management practices. Drivers often experience difficulty in finding available parking spaces, especially in busy areas such as commercial centers, offices, shopping areas, and central business districts.

The current parking system does not provide drivers with real-time information about available spaces. As a result, drivers are forced to move physically from one parking facility or street section to another while searching for space. This leads to time wastage, increased fuel consumption, traffic congestion, and frustration.

The existing system also lacks an advance reservation mechanism. Drivers cannot secure a parking space before reaching their destination. This creates uncertainty and makes parking inconvenient, especially during peak hours or in high-demand areas.

Parking attendants and operators also face operational challenges. Attendants often manage vehicle entry, exit, payment, and space allocation manually. Operators may lack centralized dashboards for monitoring parking space availability, active sessions, revenue, attendants, and reservations. This limits transparency, accountability, and data-driven decision-making.

Therefore, there was a need to design and implement a real-time parking management and space reservation system that allows drivers to view availability, reserve spaces, check in and check out digitally, make payment, and receive receipts, while also enabling parking operators and administrators to manage parking operations more efficiently.

1.4 Objectives of the Study

1.4.1 Main Objective

The main objective of the study was to design and implement a real-time parking management and space reservation system to improve parking efficiency in Kampala.

1.4.2 Specific Objectives

The specific objectives of the study were:

1. To review the current parking management system and identify gaps and limitations in the existing processes.
2. To design a real-time parking management and space reservation system that addresses the identified gaps and meets user requirements.
3. To implement the designed system using suitable web technologies.
4. To evaluate the effectiveness of the developed system in supporting parking search, reservation, check-in, check-out, payment simulation, and parking management.

1.5 Research Questions

The study was guided by the following research questions:

1. What gaps and limitations existed in the current parking management system in Kampala?
2. What system design could address the identified parking management challenges?
3. How could a real-time parking management and space reservation system be implemented using suitable web technologies?
4. How effective was the developed system in supporting parking search, reservation, check-in, check-out, payment simulation, and parking management?

1.6 Scope of the Study

The scope of the study was divided into geographical scope, content scope, and technical scope.

1.6.1 Geographical Scope

The study focused on Kampala, Uganda. Kampala was selected because it is the capital city and one of the busiest urban centers in the country. It has high vehicle movement, busy commercial areas, and parking management challenges that make it suitable for a study on real-time parking management and space reservation.

1.6.2 Content Scope

The study focused on the design and implementation of a real-time parking management and space reservation system. The system was designed to support drivers, parking attendants, parking facility operators, and system administrators.

The major system functions included user registration and login, parking facility search, real-time availability display, parking space reservation, QR-based check-in and check-out, vehicle management, fee calculation, payment simulation, receipt generation, facility management, attendant management, notifications, and administrator monitoring.

The system did not include physical parking sensors, automated gates, live license plate recognition, or full integration with live mobile money APIs. These features were considered possible areas for future improvement.

1.6.3 Technical Scope

Parka was implemented as a web-based system. The frontend was developed using React, Vite, TypeScript, and Tailwind CSS. The backend was developed using Node.js and Express. PostgreSQL was used as the database, while Socket.io was used for real-time updates and notifications. The system also used authentication and role-based access control to separate driver, attendant, operator, and administrator functions.

1.7 Significance of the Study

The study is significant to drivers, parking attendants, parking facility operators, system administrators, urban planners, researchers, and future system developers.

1.7.1 Significance to Drivers

The system helps drivers reduce the time spent searching for parking spaces by allowing them to view parking availability before arrival. It also supports advance reservation, which helps drivers secure parking spaces and plan their movement more effectively. The payment simulation and receipt generation functions also improve convenience and accountability.

1.7.2 Significance to Parking Attendants

The system supports attendants by providing digital tools for verifying reservations, checking vehicles in, checking vehicles out, and managing parking sessions. This reduces manual errors and improves the accuracy of parking records.

1.7.3 Significance to Parking Facility Operators

The system helps operators manage parking facilities, zones, spaces, attendants, reservations, sessions, and payments. It provides better visibility into occupancy and revenue, which supports improved decision-making and accountability.

1.7.4 Significance to System Administrators

The system provides administrators with tools for managing users, verifying operators, approving facilities, and monitoring system activity. This supports platform control, security, and accountability.

1.7.5 Significance to Urban Planners and Policy Makers

The study contributes to smart city and urban mobility discussions by demonstrating how digital systems can improve parking management. A system such as Parka can support better parking organization and reduce unnecessary movement caused by drivers searching for parking.

1.7.6 Significance to Researchers and Future Developers

The study contributes to academic knowledge on real-time parking management systems, reservation systems, and web-based smart city applications. It can also guide future researchers and developers who want to improve the system by integrating sensors, live mobile money APIs, artificial intelligence, or license plate recognition.

1.8 Limitations of the Study

The study had several limitations.

First, the system did not integrate physical parking sensors. Parking availability was updated through digital system actions such as reservation, check-in, and check-out rather than automatic hardware-based detection.

Second, payment was implemented as a simulation rather than a direct integration with live MTN Mobile Money or Airtel Money APIs. Live payment integration requires additional approvals, production credentials, and compliance processes.

Third, the system was tested in a controlled academic environment rather than across many real parking facilities in Kampala. This means that large-scale operational performance was not fully evaluated.

Fourth, the system depended on internet connectivity. In areas with poor connectivity, real-time updates and system access may be affected.

Fifth, the study was limited by time and financial constraints. Because it was an academic project, the implementation focused on the core functions required to demonstrate the system concept rather than advanced features such as automated gates, IoT sensors, license plate recognition, and artificial intelligence-based parking prediction.

1.9 Conceptual Framework

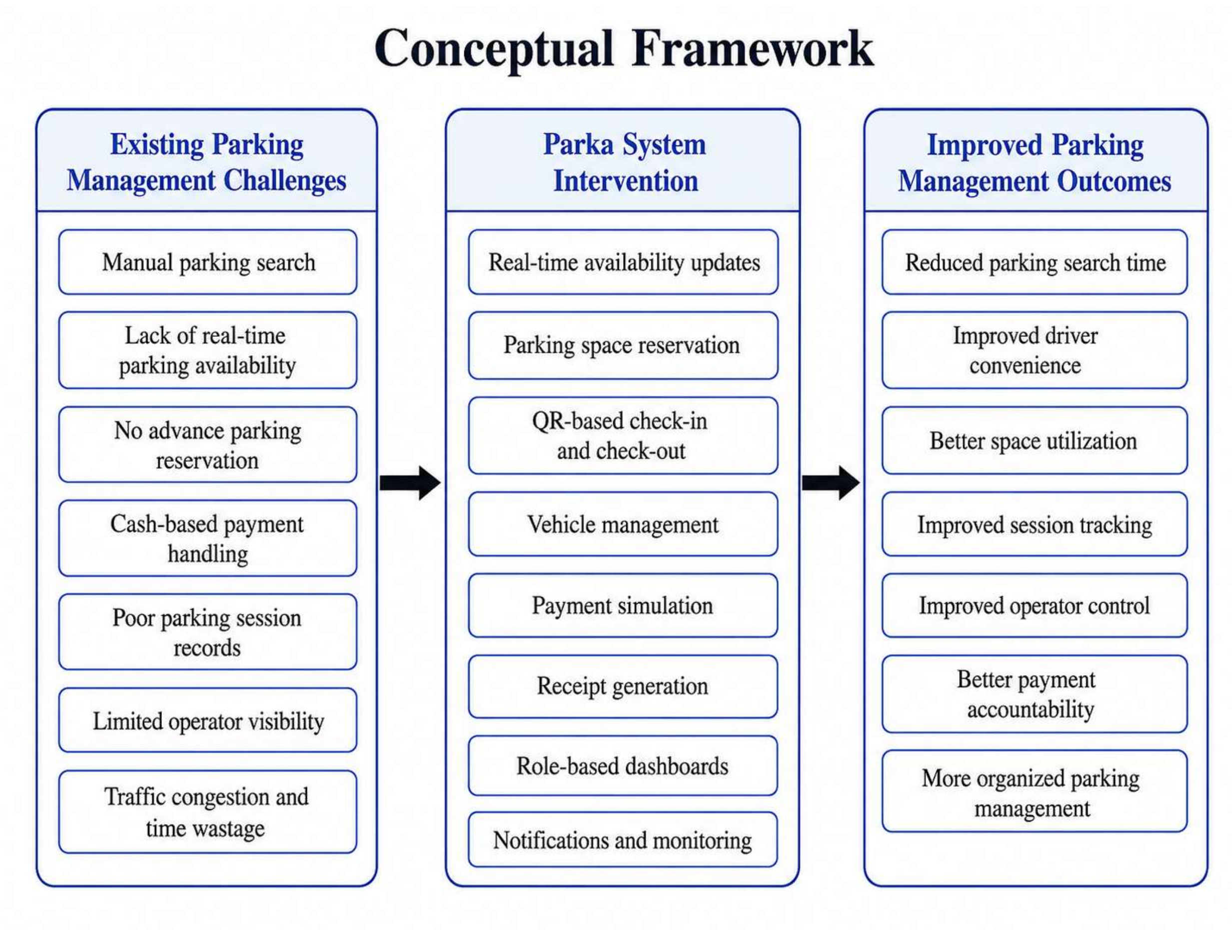
The conceptual framework shows the relationship between the existing parking challenges, the proposed system features, and the expected outcomes of the study.

The independent variables were the challenges in the existing parking system, including lack of real-time availability, manual parking search, absence of advance reservation, cash-based payments, poor record keeping, and limited operator visibility.

The intervening variable was the implementation of Parka as a real-time parking management and space reservation system. Parka introduced parking search, availability updates, reservation, QR-based check-in and check-out, payment simulation, receipt generation, role-based dashboards, notifications, and administrative monitoring.

The dependent variables were the expected outcomes, including reduced parking search time, improved convenience for drivers, better space utilization, improved parking session tracking, better operator visibility, improved accountability, and more organized parking management.

Figure 1: Conceptual Framework



1.10 Definition of Key Terms

Parking Management: The process of organizing, monitoring, controlling, and allocating parking spaces to improve accessibility and efficiency.

Real-Time Availability: The ability of a system to show updated information about available, reserved, and occupied parking spaces.

Parking Reservation: The process of booking a parking space before arriving at a parking facility.

QR Code: A machine-readable code used to store reservation or session information for quick verification.

Check-in: The process of confirming that a driver has arrived and started a parking session.

Check-out: The process of confirming that a driver is leaving the parking facility and ending the parking session.

Payment Simulation: A demonstration of a payment process without direct connection to a live financial service.

PWA: A Progressive Web Application that provides an app-like experience through a web browser.

Role-Based Access Control: A security approach where users access system functions based on their assigned roles.

1.11 Chapter Summary

This chapter introduced the study by presenting the background, statement of the problem, objectives, research questions, scope, significance, limitations, conceptual framework, and definition of key terms.

The chapter explained that Kampala's parking management challenges are linked to manual parking practices, lack of real-time availability, absence of advance reservation, poor digital records, and limited operator visibility. These challenges justified the need for Parka, a web-based real-time parking management and space reservation system.

The next chapter presents the literature review on parking management systems, smart parking systems, real-time availability systems, reservation systems, digital payments, Progressive Web Applications, and related technologies.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presents a review of literature related to parking management systems, real-time parking availability, parking reservation systems, smart parking technologies, digital payment systems, and Progressive Web Applications. The purpose of the literature review was to understand existing knowledge on parking management and identify the gaps that justified the design and implementation of Parka.

Parking management has become an important concern in urban areas because of increasing vehicle ownership, limited parking infrastructure, traffic congestion, and inefficient use of available parking spaces. In Kampala, parking management is closely connected to the wider traffic congestion problem because inefficient parking contributes to unnecessary vehicle movement, especially in busy commercial areas. A local report on Kampala traffic congestion noted that roads with fewer than three lanes should not accommodate street parking because of its effect on traffic flow (Adam Smith International, 2005, as cited in Uganda Improvement of Traffic Congestion-Flow in Kampala City, n.d.).

The Uganda Bureau of Statistics (UBOS) also emphasizes the importance of statistical information in planning, budgeting, monitoring, and decision-making in different sectors, including transport and urban development (Uganda Bureau of Statistics [UBOS], 2024).) Therefore, reviewing both global and local literature was important in understanding how digital parking systems can support better parking management in Kampala.

This chapter discusses traditional parking systems, semi-automated parking systems, smart parking systems, real-time parking availability systems, parking reservation systems, digital payment systems, Progressive Web Applications, and role-based access control. It also reviews existing parking solutions, their strengths and weaknesses, and the research gap that Parka aimed to address.

2.2 Concept of Parking Management

Parking management refers to the policies, strategies, technologies, and operational practices used to regulate and improve the use of parking spaces. It involves the planning, allocation, monitoring, pricing, and control of parking resources in order to improve accessibility, reduce congestion, and support efficient urban mobility.

According to Litman (2025), parking management includes various strategies and programs that result in more efficient use of parking resources. These strategies may include better pricing, shared parking, user information systems, regulation, enforcement, and demand management. Effective parking management improves motorist convenience, reduces parking facility costs, and supports more efficient urban transport systems (Litman, 2025).

Parking management is important because poorly managed parking can contribute to congestion, fuel wastage, air pollution, and reduced productivity. Shoup (2006) argues that drivers who cruise in search of parking increase traffic congestion and impose additional costs on other road users. In congested downtown areas, cruising for parking has been identified as a contributor to traffic flow problems because drivers continue moving while looking for vacant spaces (Shoup, 2006).

In Kampala, parking management is particularly important because of the high concentration of vehicles in the central business district and other busy urban areas. Local transport discussions have also linked poor parking control to traffic flow challenges, especially where on-street parking competes with moving traffic (Uganda Improvement of Traffic Congestion-Flow in Kampala City, n.d.). This supports the need for digital parking systems that provide real-time availability, reservation, and better coordination among drivers, attendants, operators, and administrators.

2.3 Traditional Parking Systems

Traditional parking systems are parking management approaches that depend mainly on manual processes. In such systems, parking attendants physically guide drivers, allocate spaces, collect fees, and monitor vehicle entry and exit. Records may be kept using paper receipts, manual books, verbal communication, or informal observation.

Traditional parking systems are common in many developing cities because they are easy to operate and require relatively low initial investment. However, they have several limitations. They do not provide real-time parking availability, do not support advance reservation, and do not provide reliable digital records for reporting and analysis.

In a traditional parking system, a driver usually confirms whether a space is available only after arriving at a parking location. If the facility is full, the driver must continue searching elsewhere. This leads to time wastage, fuel consumption, unnecessary movement, and frustration. Shoup (2006) explains that cruising for parking can create congestion because vehicles remain in circulation while looking for vacant spaces.

The proposal for this study also identified Kampala's current parking systems as largely manual and lacking real-time information sharing, making it difficult for drivers to know where parking spaces are available before arrival. This confirmed that traditional parking practices were not sufficient to address the parking challenges experienced in Kampala.

In Kampala, formal street parking management exists through operators such as Multiplex Limited. Multiplex publishes parking prices, sticker services, reservation-related services, and payment options for Kampala street parking (Multiplex Limited, n.d.-a). However, despite the existence of formal parking services, drivers still require better real-time visibility, digital reservation support, and integrated parking management.

2.4 Semi-Automated Parking Systems

Semi-automated parking systems introduce some level of technology while still depending partly on manual operations. Examples include ticket-based parking systems, barrier-controlled parking, printed receipts, entry and exit machines, and basic computerized payment terminals.

In semi-automated systems, a driver may receive a ticket upon entry and pay based on the duration of stay. This improves record keeping compared to fully manual systems because entry and exit times may be captured. Some systems may also use payment machines or attendant-operated computers.

Semi-automated systems improve revenue tracking and reduce some manual errors. However, they may still fail to provide real-time availability information to drivers before arrival. Many of them also do not support advance reservation or integration with mobile-based platforms. Therefore, although semi-automated systems are better than fully manual systems, they still do not fully solve the problem of drivers physically searching for available spaces.

In Kampala, earlier street parking modernization efforts introduced automated or machine-supported payment options, including smart card or coin-based payment machines, showing that the city has attempted to improve parking operations through technology (Uganda Radio Network, 2012).) However, such approaches still require stronger integration with real-time availability, reservation, digital verification, and role-based management.

2.5 Smart Parking Systems

Smart parking systems are technology-based parking solutions that use digital tools to improve parking availability, reservation, monitoring, payment, enforcement, and reporting. These systems may use mobile applications, web platforms, sensors, cameras, GPS, maps, databases, cloud platforms, and real-time communication technologies.

Smart parking systems aim to reduce the time drivers spend searching for parking spaces by providing accurate information about availability. They also help parking operators monitor occupancy, manage revenue, improve space utilization, and analyze parking usage patterns.

Geng and Cassandras (2013) proposed a smart parking system that assigns and reserves an optimal parking space based on the driver’s cost function, including proximity to destination and parking cost. Their study demonstrates that parking reservation and resource allocation can improve parking efficiency in urban environments.

Barone et al. (2014) also proposed an architecture for parking management in smart cities. Their work shows that smart parking systems can provide drivers with information about parking availability and support more efficient use of urban parking resources (Barone et al., 2014).

Smart parking systems may include real-time parking availability, advance reservation, digital payment, navigation to parking facilities, automated check-in and check-out, operator dashboards, notifications, analytics, and reporting. These features make smart parking systems more effective than traditional systems. However, many smart parking solutions require expensive infrastructure such as sensors, automated gates, and camera systems. This may limit adoption in developing cities where financial and technical resources are constrained.

2.6 Real-Time Parking Availability Systems

Real-time parking availability systems provide updated information about available and occupied parking spaces. This information can be displayed through mobile applications, websites, digital signboards, or dashboards used by parking operators.

Real-time availability is important because it helps drivers make informed decisions before arriving at a parking facility. Instead of moving from one location to another physically, a driver can check available spaces digitally and choose a suitable facility.

In smart parking systems, real-time availability may be achieved through sensors, cameras, ticketing systems, attendant updates, or software events such as reservations, check-ins, and check-outs. Biyik et al. (2021) explain that smart parking systems use different technologies and architectures to detect occupancy and provide parking information to users.

Geng and Cassandras (2013) also show that assigning and reserving spaces using smart parking logic can reduce the inefficiencies associated with uncontrolled parking processes. This supports the importance of real-time parking information in reducing search time and improving utilization.

For Parka, real-time availability was implemented through digital updates from reservations, check-ins, and check-outs. When a driver reserves a space, the system updates availability. When a vehicle checks in, the space becomes occupied. When the vehicle checks out, the space becomes available again. This approach provides real-time parking information without requiring expensive physical sensors.

2.7 Parking Reservation Systems

Parking reservation systems allow drivers to book parking spaces before arriving at a parking facility. This reduces uncertainty and ensures that a driver has a reserved space at the destination.

Reservation systems are useful in busy urban areas where parking demand is high. They allow drivers to plan ahead and avoid wasting time searching for parking. They also help parking operators forecast demand and manage space allocation more effectively.

A parking reservation process usually involves selecting a facility, choosing a time or duration, selecting a vehicle, confirming the booking, and receiving a reservation code or QR token. The code is then used during check-in to verify the reservation.

Geng and Cassandras (2013) emphasize that smart parking systems can reserve parking resources based on driver needs and system-level efficiency. Barone et al. (2014) also describe intelligent parking systems as solutions that can provide availability information and allow users to reserve convenient spaces before arrival.

In Kampala, reservation-based parking is relevant because existing operators already refer to reservation-related parking areas and payment services (Multiplex Limited, n.d.-a). However, there is still a need for a complete digital system that connects drivers, attendants, operators, and administrators in real time.

Parka applied this concept by allowing drivers to reserve parking spaces in advance and receive reservation confirmation through a code or QR token. The reservation was then verified by the parking attendant during check-in.

2.8 Digital Payment in Parking Management

Digital payment refers to the use of electronic methods to pay for parking services. These may include mobile money, debit cards, credit cards, online wallets, or payment gateways.

In traditional parking systems, payments are often made in cash. Cash payments are simple but may create problems such as poor accountability, lack of receipts, revenue leakage, and difficulty in preparing financial reports. Digital payment improves transparency because each transaction can be recorded and linked to a parking session. It also improves convenience because drivers do not need to carry cash. For operators, digital payment supports better revenue tracking, reporting, and auditing.

In Kampala, Multiplex Limited publishes parking prices and indicates that parking can be paid through various methods. The existence of mobile and digital payment options for parking in Kampala shows that digital payment is relevant to the local parking environment.

For Parka, payment was implemented as a simulator to demonstrate a mobile money-style parking payment process. The system calculates the parking fee, simulates payment confirmation, records the transaction, and generates a receipt. This provides a functional model for future integration with live MTN Mobile Money or Airtel Money APIs.

2.9 Progressive Web Applications in Parking Systems

A Progressive Web Application is a web application that provides an app-like experience through a web browser. PWAs are designed to work across different devices, including mobile phones, tablets, and desktop computers. They can support responsive design, offline caching, fast loading, and installation-like behavior on some devices.

PWAs are useful for parking systems because drivers and attendants may not want to download a native mobile application before using the service. A PWA can be accessed through a link, making it easier to reach users quickly. PWAs are also cost-effective because one application can serve users on different platforms. Instead of developing separate Android, iOS, and desktop applications, a web-based PWA can support multiple devices using one codebase.

For Parka, a web-based approach was suitable because the system needed to serve drivers, attendants, operators, and administrators. Drivers and attendants could access the system on mobile phones, while operators and administrators could use larger screens for dashboards and management functions. The final Parka documentation confirms that the system was developed as a React-based web platform with separate portals for drivers, attendants, operators, and administrators.

2.10 Role-Based Access Control in Parking Management Systems

Role-based access control is a security approach where system permissions are assigned according to user roles. In a parking management system, different users perform different tasks. Therefore, they should not all have the same level of access.

For example, a driver should be allowed to search for parking, reserve spaces, manage vehicles, pay, and view receipts. A parking attendant should be allowed to verify reservations, check vehicles in, and check vehicles out. A facility operator should be allowed to manage facilities, spaces, attendants, and pricing. A system administrator should be allowed to manage users, approve facilities, and monitor the platform.

Role-based access improves system security and organization. It prevents unauthorized users from accessing sensitive functions. It also makes the system easier to use because each user only sees the features relevant to their responsibilities.

Parka applied role-based access control by providing separate portals for drivers, attendants, operators, and administrators. The system documentation identifies these four user journeys as central to the design of Parka.

2.11 Existing Parking Solutions

Several parking management solutions have been developed in different parts of the world. These solutions vary depending on the technology used, cost of implementation, level of automation, and target users.

The major categories of existing parking solutions include sensor-based parking systems, mobile-based parking applications, smart parking guidance systems, and license plate recognition systems.

2.11.1 Sensor-Based Parking Systems

Sensor-based parking systems use physical sensors installed in parking spaces to detect whether a space is occupied or available. The sensors send data to a central system, which updates parking availability.

These systems provide high accuracy because they directly detect vehicle presence. They are useful in controlled facilities such as shopping malls, airports, hospitals, and multi-storey parking buildings. Biyik et al. (2021) explain that smart parking systems use different technologies such as sensors, cameras, and connected platforms to collect and communicate occupancy data.

However, sensor-based systems are expensive to install and maintain. They require hardware, power, connectivity, and technical maintenance. In developing urban contexts, these costs may be difficult for small parking operators to afford.

2.11.2 Mobile-Based Parking Applications

Mobile-based parking applications allow users to search for parking, view available spaces, reserve slots, and sometimes pay digitally. These applications improve convenience because users can access parking information from their phones.

In Uganda, the Plex Parking application by Multiplex Limited is an example of a local parking-related mobile application. The application description indicates that users can monitor parking sessions, pay parking fees and fines, find parking using location, and get directions to parking spaces

The main advantage of mobile parking applications is user convenience. However, their effectiveness depends on user adoption, internet access, and participation by parking operators. In some cases, they may not provide accurate availability if they are not connected to reliable live parking data.

2.11.3 Smart Parking Guidance Systems

Smart parking guidance systems use digital signs, indicators, or display boards to direct drivers to available parking spaces. These systems are common in large parking facilities such as shopping malls, airports, and multi-level parking buildings.

They improve navigation within a facility and reduce internal searching. However, they are usually limited to specific physical locations and may not help drivers before they reach the facility. These systems also require installation of digital boards, sensors, controllers, and supporting infrastructure, which may increase implementation costs.

2.11.4 License Plate Recognition Systems

License plate recognition systems use cameras and image processing technologies to identify vehicles based on their number plates. These systems can automate entry, exit, billing, and security monitoring.

License plate recognition improves automation and can reduce the need for manual verification. However, it is expensive to implement and may raise privacy concerns because it involves capturing and processing vehicle identity data.

In the Ugandan context, recent research has explored intelligent traffic surveillance using vehicle detection and license plate recognition for traffic management and automated enforcement (Mugizi et al., 2026). Although such technologies are promising, they require cameras, image processing models, reliable infrastructure, and proper data protection measures. For an academic parking management project, a QR-based check-in and check-out approach was more affordable and practical.

2.12 Strengths and Weaknesses of Existing Parking Systems

Existing parking solutions have different strengths and weaknesses. Traditional systems are easy to operate but lack real-time information and digital records. Sensor-based systems provide accurate availability but are expensive. Mobile parking applications improve convenience but depend on adoption and accurate data. License plate recognition systems improve automation but are costly and may raise privacy concerns.

Table 2.1: Strengths and Weaknesses of Existing Parking Systems

System Type	Strengths	Weaknesses
Traditional parking systems	Simple to operate and low setup cost	Manual, inefficient, no real-time information, poor records
Semi-automated systems	Better than manual systems and can support ticketing	Limited real-time availability and reservation support
Sensor-based systems	Accurate real-time space detection	Expensive installation and maintenance
Mobile parking applications	Convenient and accessible to users	Depends on user adoption and accurate operator data
Smart guidance systems	Useful within large parking facilities	Limited to specific locations and requires infrastructure
License plate recognition systems	Improves automation and security	High cost, privacy concerns, and technical complexity

2.13 Comparative Review of Existing Systems and Parka

A comparative review helps show how Parka differs from existing parking management approaches. Unlike traditional systems, Parka provides digital parking search, reservation, and session tracking. Unlike sensor-based systems, Parka does not require expensive physical sensors. Unlike basic mobile parking apps, Parka includes role-based portals for drivers, attendants, operators, and administrators.

Table 2.2: Comparative Review of Existing Systems and Parka

Feature	Traditional System	Sensor-Based Smart System	Mobile Parking App	Parka
Real-time availability	No	Yes	Sometimes	Yes
Advance reservation	No	Sometimes	Yes	Yes

Feature	Traditional System	Sensor-Based Smart System	Mobile Parking App	Parka
QR check-in/check-out	No	Sometimes	Sometimes	Yes
Digital payment support	No	Yes	Yes	Simulated for academic scope
Operator dashboard	No	Yes	Sometimes	Yes
Attendant portal	No	Sometimes	Rare	Yes
Administrator portal	No	Yes	Sometimes	Yes
Requires expensive sensors	No	Yes	Sometimes	No
Suitable for low-cost implementation	Yes, but limited	No	Moderate	Yes
Digital records and receipts	No	Yes	Yes	Yes

This comparison shows that Parka was designed to provide key smart parking features while avoiding heavy dependence on expensive physical infrastructure.

2.14 Benefits of Real-Time Parking Management Systems

Real-time parking management systems provide several benefits to drivers, parking operators, and urban authorities.

For drivers, these systems reduce uncertainty by showing where parking spaces are available. They also reduce the time spent searching for parking and improve convenience through advance reservation and digital payment.

For parking operators, real-time systems improve monitoring of parking spaces, reservations, revenue, and attendants. They also support better decision-making through digital records and reports.

For urban authorities, real-time parking systems can contribute to reduced congestion by minimizing unnecessary vehicle movement. Shoup (2006) shows that cruising for parking contributes to traffic congestion, while smart parking studies show that reservation and information systems can reduce search time and improve parking resource use (Geng & Cassandras, 2013; Shoup, 2006).

In Kampala, a real-time parking system can help improve urban mobility by reducing physical searching for parking and improving utilization of available parking facilities. This is consistent with local transport concerns that better planning of roads, parking, and public transport is required to improve traffic flow in Kampala (Uganda Improvement of Traffic Congestion-Flow in Kampala City, n.d.).

2.15 Challenges of Implementing Smart Parking Systems

Despite their benefits, smart parking systems face several implementation challenges. One major challenge is cost. Systems that depend on sensors, automated gates, cameras, and digital signboards can be expensive to install and maintain.

Another challenge is internet connectivity. Real-time parking systems depend on reliable connectivity to update and transmit information. Poor internet access may affect system performance, especially in some urban or peri-urban areas.

A third challenge is user adoption. Drivers, attendants, and operators must be willing to use the system. If users continue relying only on manual methods, the effectiveness of the system may be reduced.

Another challenge is data accuracy. If parking space status is not updated correctly, users may receive incorrect information. Therefore, the system must ensure that reservations, check-ins, and check-outs update space availability accurately.

Security and privacy are also important concerns. Parking systems handle user accounts, vehicle details, payment records, and location-related information. These must be protected from unauthorized access. Parka addressed some of these challenges by using a web-based approach, role-based access control, password hashing, token-based authentication, and real-time updates based on system actions.

2.16 Research Gap

The literature reviewed shows that parking management has evolved from traditional manual systems to smart technology-based systems. Existing smart parking solutions provide useful features such as real-time availability, digital payment, reservation, and automated monitoring.

However, several gaps remain, especially in the context of developing cities such as Kampala. Many smart parking systems depend on expensive infrastructure such as sensors, cameras, automated gates, and digital signboards. These technologies may not be affordable for many small and medium parking operators.

Traditional parking systems, on the other hand, are cheaper but lack real-time information, advance reservation, digital payment records, and operator dashboards. Mobile-based parking applications provide convenience but may not always include complete role-based management for drivers, attendants, operators, and administrators.

In Kampala, formal parking services, pricing, enforcement, and mobile payment options exist through operators such as Multiplex Limited, but there remains a need for stronger real-time visibility, reservation support, QR-based verification, and centralized management. Therefore, there was a need for a cost-effective, web-based parking management and space reservation system that could provide real-time availability, advance reservation, QR-based check-in and check-out, payment simulation, receipt generation, and facility management without relying heavily on expensive hardware.

Parka was designed to address this gap by providing a web-based solution suitable for Kampala’s parking context. It focused on digital coordination, reservation, session tracking, role-based dashboards, and real-time updates while leaving room for future integration with sensors, live mobile money APIs, and other advanced smart city technologies.

2.17 Chapter Summary

This chapter reviewed literature related to parking management systems, traditional parking systems, semi-automated systems, smart parking systems, real-time parking availability, reservation systems, digital payments, Progressive Web Applications, and role-based access control.

The review showed that traditional parking systems are simple but inefficient because they lack real-time availability, advance reservation, digital records, and centralized monitoring. Smart parking systems provide better functionality, but many of them depend on expensive infrastructure such as sensors, cameras, and automated gates.

The chapter also reviewed existing parking solutions, including sensor-based systems, mobile applications, smart parking guidance systems, and license plate recognition systems. Their strengths and weaknesses were compared with Parka.

The research gap identified was the need for a cost-effective, web-based, real-time parking management and reservation system suitable for Kampala. Parka was designed and implemented to address this gap by supporting parking search, availability updates, reservation, QR-based check-in and check-out, payment simulation, receipt generation, operator dashboards, and administrator monitoring.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter presents the methodology that was used in carrying out the study on the design and implementation of **Parka**, a real-time parking management and space reservation system for Kampala. It explains the research design, study population, sampling techniques, data collection methods, data analysis techniques, system development methodology, development tools and technologies, system testing methods, and ethical considerations.

The purpose of the methodology was to guide the study in identifying the existing parking management challenges, gathering user and system requirements, designing a suitable solution, implementing the system, and evaluating whether the developed system addressed the identified problems. Since the study involved both understanding an existing problem and developing a working software solution, it combined research methods with system development methods.

The study followed a case study approach focusing on parking management challenges in Kampala. It also applied a developmental approach because the final output of the study was a working web-based real-time parking management and space reservation system. The proposed methodology in the earlier proposal emphasized case study research, descriptive analysis, Agile development, and system testing, which were followed during the final implementation of Parka.

3.2 Research Design

The study adopted a **case study research design**, supported by descriptive and developmental approaches. The case study design was suitable because the research focused on a specific real-world problem: parking inefficiency in Kampala. Kampala was selected because it experiences high vehicle movement, limited organized parking infrastructure, and parking congestion, especially in busy commercial and urban areas.

The descriptive approach was used to understand the existing parking management practices, the challenges faced by drivers, the limitations experienced by parking attendants and operators, and the need for a digital parking management solution. This helped the researcher to describe the current situation and identify gaps in the existing system.

The developmental approach was used because the study involved the design and implementation of a software system. After identifying the weaknesses of the existing system, Parka was designed and developed as a practical solution to support real-time parking availability, advance reservation, QR-based check-in and check-out, payment simulation, receipt generation, and digital facility management.

Therefore, the research design was appropriate because it allowed the study to investigate the problem, gather requirements, design the solution, implement the system, and evaluate its performance.

3.3 Study Population

The study population consisted of people and groups directly affected by parking management challenges in Kampala. These included drivers, parking attendants, parking facility operators, and urban road users. These groups were selected because they interact with parking services regularly and experience the effects of inefficient parking management.

Drivers were included because they are the main users who search for parking spaces and experience challenges such as time wastage, fuel consumption, and uncertainty of parking availability.

Parking attendants were included because they participate in the daily management of parking spaces, vehicle entry, vehicle exit, payment collection, and physical monitoring of parking areas.

Parking facility operators were included because they manage parking facilities and require tools for monitoring occupancy, revenue, attendants, reservations, and parking sessions.

Urban road users were also considered because inefficient parking affects traffic flow and contributes to congestion in busy areas.

Table 3.1: Study Population

Category	Relevance to the Study
Drivers	They experience difficulty in finding available parking spaces
Parking attendants	They manage vehicle entry, exit, and parking sessions manually
Parking facility operators	They manage parking facilities, revenue, spaces, and attendants
Urban road users	They are affected by traffic congestion caused by parking inefficiencies

3.4 Sampling Techniques

The study used **purposive sampling** and **convenience sampling**.

Purposive sampling was used to select respondents who had direct experience with parking challenges. These included drivers who frequently use parking facilities in Kampala, parking attendants who manage parking spaces, and parking operators involved in facility management. This technique was useful because it allowed the researcher to collect information from people who understood the parking problem.

Convenience sampling was also used because of time and accessibility constraints. Some respondents were selected based on their availability and willingness to participate in the study. This was practical for an academic project because it allowed data to be collected within the available time and resources.

The combination of purposive and convenience sampling helped the researcher obtain relevant information from respondents who were directly or indirectly involved in parking activities.

3.5 Data Collection Methods

The study used both primary and secondary data collection methods. Primary data was collected directly from respondents, while secondary data was obtained from existing literature, reports, journals, and online sources related to parking management systems and smart parking solutions.

3.5.1 Primary Data Collection

Primary data was collected using questionnaires, interviews, and observation.

Questionnaires

Questionnaires were used to collect information from drivers and other road users. The questionnaires focused on the challenges drivers face when searching for parking spaces, the time spent looking for parking, the availability of parking information, willingness to use a digital parking system, and expectations from a parking reservation platform.

The questionnaire method was useful because it allowed the researcher to gather structured responses that could be analyzed using frequencies and percentages.

Interviews

Interviews were used to gather information from parking attendants and parking facility operators. The interviews focused on current parking management practices, methods used to monitor parking spaces, challenges in handling vehicle entry and exit, payment handling, record keeping, and the need for digital tools.

The interview method helped the researcher understand practical parking operations from the perspective of people who manage parking facilities.

Observation

Observation was used to understand how parking is managed in real-life settings. The researcher observed how drivers search for spaces, how attendants guide vehicles, how payments are handled, and how parking spaces are monitored manually.

Observation helped confirm some of the weaknesses of the existing system, including lack of real-time parking information, manual checking of space availability, and informal record keeping.

3.5.2 Secondary Data Collection

Secondary data was collected from academic journals, books, government reports, online articles, and existing studies on parking management, smart parking systems, real-time information systems, and urban traffic congestion.

This data helped the researcher understand existing parking technologies, the evolution of parking management systems, benefits of smart parking, limitations of current solutions, and the research gap that Parka aimed to address.

The proposal had identified secondary sources such as academic journals, government reports, and existing parking studies as important sources for understanding global and local parking management issues.

3.6 Data Analysis Techniques

The collected data was analyzed using both quantitative and qualitative analysis techniques.

3.6.1 Quantitative Data Analysis

Quantitative data from questionnaires was analyzed using simple statistical methods such as frequencies, percentages, and tabular presentation. This helped in summarizing responses from drivers and other participants.

For example, questionnaire responses were used to identify how many respondents experienced difficulty in finding parking, how many supported the idea of a digital parking system, and how many preferred advance parking reservation.

The results were presented using tables and, where necessary, charts.

3.6.2 Qualitative Data Analysis

Qualitative data from interviews and observation was analyzed using thematic analysis. This involved identifying common themes from the responses and observations. The major themes included lack of real-time parking information, manual parking management, poor record keeping, lack of advance reservation, payment challenges, and need for digital monitoring.

Thematic analysis was useful because it allowed the researcher to interpret the views of parking attendants and operators in relation to the parking management problem.

3.7 System Development Methodology

The system was developed using the **Agile software development methodology**. Agile was selected because it supports iterative development, continuous testing, user feedback, and progressive improvement of system features.

The Agile methodology was suitable for this project because the system required several functional modules, including authentication, facility management, parking space management, reservation, QR check-in and check-out, payment simulation, notifications, and administration. These modules were developed and improved in stages.

The use of Agile allowed the researcher to begin with core features and then improve the system gradually. For example, the system first required basic user registration and login, followed by role-based dashboards, facility management, reservation, QR verification, payment simulation, and real-time updates.

The proposal also identified Agile as the preferred system development methodology because it supports incremental development, flexibility, feedback, and early delivery of a working prototype.

3.7.1 Phases of System Development

The development of Parka followed several phases: requirements analysis, system design, implementation, testing, and deployment preparation.

Requirements Analysis

During this phase, the researcher analyzed the existing parking system and identified user requirements. The main problems identified included lack of real-time parking information, absence of advance reservation, manual record keeping, cash-based payment, and limited operator visibility.

The requirements gathered during this phase guided the functional and non-functional requirements of the system.

System Design

In this phase, the structure of the system was designed. The system design included the architecture, user roles, database model, workflows, use case diagram, context diagram, reservation workflow, check-in and check-out workflow, and payment workflow.

The system was designed as a three-tier web application consisting of a frontend, backend, and database.

Implementation

During implementation, the system was developed as a working web-based application. The frontend was implemented using React, while the backend was implemented using Node.js and Express. PostgreSQL was used as the database, and Socket.io was used to support real-time communication.

The implemented modules included authentication, driver portal, attendant portal, operator portal, administrator portal, reservation engine, QR check-in and check-out, payment simulation, receipt generation, and notifications.

Testing

Testing was conducted to verify whether the system worked according to the requirements. The system was tested for functionality, usability, performance, security, reliability, and data integrity.

Deployment Preparation

The system was prepared for deployment as a web-based application. The frontend and backend were organized separately, and the database was configured to support system operations.

3.8 Development Tools and Technologies

The system was developed using modern web technologies. These tools were selected because they support scalability, responsiveness, real-time updates, and database-driven web applications.

Table 3.2: Development Tools and Technologies

Tool / Technology	**Purpose**
React	Used to develop the frontend user interface
Vite	Used as the frontend build tool
TypeScript	Used to improve code structure and reliability
Tailwind CSS	Used for styling and responsive user interface design
Node.js	Used to run the backend server
Express.js	Used to build RESTful backend APIs
PostgreSQL	Used as the relational database

Tool / Technology	Purpose
Socket.io	Used for real-time updates and notifications
JWT	Used for secure authentication and protected routes
bcrypt	Used for password hashing
Helmet	Used to improve API security
CORS	Used to control frontend-backend communication
Rate limiting	Used to reduce abuse of API endpoints

3.9 System Architecture Method

Parka was implemented using a three-tier architecture. The architecture separated the system into the presentation layer, application logic layer, and data layer.

The presentation layer consisted of the React frontend used by drivers, attendants, operators, and administrators. The application logic layer consisted of the Node.js and Express backend, which handled business logic and API requests. The data layer consisted of the PostgreSQL database, which stored users, vehicles, facilities, parking spaces, reservations, sessions, payments, receipts, and notifications.

This architecture was used because it improves system organization, security, scalability, and maintainability.

3.10 System Testing Methods

System testing was conducted to ensure that Parka met its functional and non-functional requirements. The testing methods included functional testing, usability testing, performance testing, security testing, and reliability testing.

3.10.1 Functional Testing

Functional testing was used to verify whether each system feature worked as expected. The tested features included user registration, login, parking facility search, reservation, QR generation, check-in, check-out, fee calculation, payment simulation, receipt generation, notifications, and role-based dashboards.

3.10.2 Usability Testing

Usability testing was conducted to determine whether the system was easy to use. The focus was on navigation, interface clarity, mobile responsiveness, ease of reservation, ease of QR verification, and dashboard usability.

This was important because the system was expected to serve different users with different levels of technical experience.

3.10.3 Performance Testing

Performance testing was used to assess how quickly the system responded to user actions. The tested areas included page loading, login, parking search, reservation processing, QR verification, payment simulation, and real-time updates.

The aim was to ensure that the system could respond within an acceptable time during normal use.

3.10.4 Security Testing

Security testing was conducted to verify whether the system protected user accounts and restricted access to authorized users. The testing focused on login protection, password hashing, JWT authentication, protected routes, role-based access control, and API security middleware.

Security testing was important because the system manages user accounts, vehicle information, reservations, payments, and administrative controls.

3.10.5 Reliability and Data Integrity Testing

Reliability and data integrity testing was conducted to ensure that the system maintained accurate records during reservation, check-in, checkout, and payment. The testing focused on whether parking space statuses were updated correctly and whether the system prevented inconsistent records.

For example, when a space was reserved, it was expected to change from available to reserved. When a driver checked in, the space was expected to change to occupied. When the driver checked out and payment was completed, the space was expected to return to available.

3.11 Ethical Considerations

The study followed ethical principles during data collection, system design, and reporting. Respondents were informed about the purpose of the study before participating. Participation was voluntary, and respondents were not forced to provide information.

Confidentiality was maintained by ensuring that personal information collected during the study was used only for academic purposes. The study avoided exposing sensitive respondent information in the final report.

During system development, user data protection was also considered. Passwords were hashed, access to system functions was controlled through authentication, and user roles were used to limit access to specific system areas.

The ethical considerations followed in this study included informed consent, confidentiality, voluntary participation, and responsible handling of information.

3.12 Chapter Summary

This chapter presented the methodology used in the study. It explained the research design, study population, sampling techniques, data collection methods, data analysis techniques, system development methodology, development tools, testing methods, and ethical considerations.

The study adopted a case study design supported by descriptive and developmental approaches. Data was collected using questionnaires, interviews, observation, and secondary sources.

Quantitative data was analyzed using frequencies and percentages, while qualitative data was analyzed thematically.

The system was developed using Agile methodology because it supported iterative development, continuous testing, and progressive improvement. The implementation used React, Node.js, Express, PostgreSQL, Socket.io, JWT authentication, password hashing, and other supporting technologies.

The chapter also explained the testing methods used to evaluate the system, including functional, usability, performance, security, reliability, and data integrity testing. Ethical considerations such as informed consent, confidentiality, voluntary participation, and data protection were observed throughout the study.

CHAPTER FOUR: SYSTEM ANALYSIS, DESIGN & IMPLEMENTATION

4.1 Introduction

This chapter presents the system analysis, design, and implementation of Parka, a real-time parking management and space reservation system developed to address parking inefficiencies in Kampala. The chapter explains the existing parking management practices, identifies the strengths and weaknesses of the current system, presents the requirements of the proposed system, and describes how the developed system was designed and implemented.

4.2 Analysis of the Existing System

The existing parking management system in Kampala is largely manual and depends on physical searching, parking attendants, cash-based payments, and informal record keeping. In many busy areas of Kampala, drivers are required to move from one parking facility or street section to another while searching for available parking spaces. This process is inconvenient because drivers do not have prior information about where parking is available before reaching their destination.

In the current system, parking attendants play a central role in directing vehicles, identifying available spaces, collecting parking fees, and monitoring vehicle entry and exit. Although attendants help in managing parking physically, the process lacks digital coordination. Parking availability is not updated in real time, drivers cannot reserve spaces in advance, and parking operators do not have reliable dashboards for monitoring occupancy, reservations, revenue, or facility performance.

The existing system also depends heavily on manual judgement. A driver only confirms whether a space is available after reaching the parking location. If the space is unavailable, the driver must continue searching elsewhere. This results in unnecessary movement, increased fuel consumption, time wastage, traffic congestion, and frustration among drivers.

It was noted that traditional parking systems in Kampala were mainly manual and lacked real-time information sharing. It was also noted that drivers often spent considerable time searching for parking spaces, especially in central business districts and other high-demand urban areas.

This process is simple but inefficient because it does not provide real-time parking visibility, advance booking, automated session tracking, digital payment records, or centralized facility management.

4.3 Strengths of the Existing System

Although the existing parking management system has several limitations, it also has some strengths that were considered during the design of Parka. These strengths show why the manual system has continued to operate despite its weaknesses.

First, the existing system is simple to understand and operate. Drivers and parking attendants are already familiar with the process of physically locating a parking space, receiving guidance from attendants, paying parking fees, and leaving the parking area after use. This simplicity makes the system easy to use, especially for people with limited experience using digital platforms.

Second, the existing system requires low initial technology investment. Since most parking operations are carried out manually, operators do not need complex software, servers, internet connectivity, automated barriers, sensors, or digital dashboards in order to run basic parking activities.

Third, the presence of parking attendants provides human support to drivers. Attendants help guide vehicles, identify available spaces, explain parking rules, and respond to immediate challenges that drivers may face when entering or leaving a parking facility.

Fourth, the existing system is flexible in handling different parking situations. For example, attendants can manually guide drivers to alternative spaces, allow walk-in vehicles, and communicate directly with drivers when spaces are full or when there are special parking conditions.

Lastly, the existing system already has an established payment culture. Drivers are used to paying for parking services through cash, mobile money, stickers, or other arrangements. This existing payment habit made it easier for Parka to introduce payment simulation and provide a foundation for future live digital payment integration.

Table 4.1: Strengths of the Existing Parking Management System

Strength	**Benefit to Parking Operations**
Simple to operate	Users can easily understand the parking process
Low initial technology cost	Operators can run basic parking activities without expensive digital systems
Human assistance from attendants	Drivers receive direct support during parking
Flexibility in handling parking situations	Attendants can respond to different parking conditions manually
Existing payment culture	Drivers are already familiar with paying for parking services

4.4 Weaknesses of the Existing System

The analysis of the existing parking management system revealed several weaknesses that justified the development of Parka.

First, the existing system lacks real-time parking availability information. Drivers are only able to know whether a parking space is available after reaching a parking location. This results in unnecessary movement from one facility to another and increases the time spent searching for parking.

Second, the current system does not support advance reservation. A driver cannot book a space before arriving at a destination. This creates uncertainty, especially in busy commercial areas where parking demand is high.

Third, the system depends heavily on manual attendants. Although attendants support day-to-day parking operations, manual handling may lead to delays, errors, inconsistent record keeping, and limited accountability.

Fourth, the payment process is largely cash-based. Cash handling limits transparency, makes it difficult to track revenue accurately, and reduces the ability of operators to generate reliable receipts or transaction reports.

Fifth, parking operators lack centralized digital dashboards for monitoring facility occupancy, active sessions, reservations, attendants, payments, and revenue. This makes it difficult to make informed operational decisions.

Lastly, the existing system does not provide sufficient digital records for auditing and future analysis. Without reliable data, parking operators cannot easily identify usage patterns, peak hours, underutilized spaces, or revenue trends.

Table 4.2: Weaknesses of the Existing Parking Management System

Weakness	Effect on Parking Management
Lack of real-time availability information	Drivers waste time searching for parking spaces
Absence of advance reservation	Drivers cannot secure parking before arrival
Heavy dependence on manual attendants	Increases chances of delay, error, and inconsistent service
Cash-based payment process	Limits transparency and weakens revenue tracking
Poor session tracking	Makes it difficult to monitor vehicle entry, stay duration, and exit
Lack of centralized operator dashboard	Operators cannot effectively monitor occupancy and performance
Limited digital records	Reduces accountability, reporting, and future analysis

4.5 Requirements of the Proposed System

The proposed system was designed to address the limitations of the existing parking management system. The requirements were derived from the parking challenges identified in Kampala and from the need to provide a digital platform that improves parking search, reservation, monitoring, and payment handling.

The system requirements were categorized into functional requirements, non-functional requirements, and user requirements.

4.5.1 Functional Requirements

Functional requirements describe the specific operations that the system was expected to perform. Parka was designed to support the main activities of drivers, parking attendants, parking facility operators, and system administrators.

Table 4.3: Functional Requirements of the Proposed System

Requirement	**Description**
User registration and login	The system allows users to create accounts and log in securely.
Role-based access control	The system provides different dashboards and permissions for drivers, attendants, operators, and administrators.
Parking facility search	Drivers can search and view available parking facilities.
Real-time availability display	The system displays updated parking space availability.
Parking reservation	Drivers can reserve parking spaces before arrival.
Reservation code or QR generation	The system generates a reservation code or QR token after successful booking.

Requirement	Description
Vehicle management	Drivers can add and manage their vehicle information.
QR-based check-in	Attendants verify reservations and check drivers in using QR codes or reservation codes.
QR-based check-out	Attendants verify exiting vehicles and close active parking sessions.
Parking fee calculation	The system calculates parking charges based on duration and pricing rules.
Payment simulation	The system simulates mobile money-style payment for parking charges.
Receipt generation	The system generates a receipt after successful payment.
Facility management	Operators can create and manage parking facilities.
Zone and space management	Operators can define parking zones and individual spaces.
Attendant management	Operators can assign attendants to parking facilities.
Notifications	The system sends reservation, check-in, check-out, payment, and session notifications.
Administrative monitoring	Administrators manage users, verify operators, approve facilities, and monitor system activity.

These requirements correspond to the implemented Parka system, which was structured as a digital parking marketplace connecting drivers, attendants, operators, and administrators.

4.5.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes expected from the system. These requirements were important because the system needed to be usable, secure, reliable, scalable, and maintainable.

Table 4.4: Non-Functional Requirements of the Proposed System

Requirement	**Description**
Usability	The system should be easy to use and should provide clear navigation for all user roles.
Security	User data, passwords, and protected functions should be secured using authentication and authorization mechanisms.
Reliability	The system should consistently manage reservations, check-ins, check-outs, payments, and space status.
Performance	The system should respond quickly when users search, reserve, check in, check out, or make payments.
Scalability	The system should support future expansion to more parking facilities, users, and locations.
Maintainability	The system should be modular and easy to update or improve.
Data integrity	The system should prevent duplicate reservations, inaccurate space status, and inconsistent records.
Compatibility	The system should work on both mobile and desktop devices through a web browser.
Availability	The system should be accessible as a web-based platform whenever internet

Requirement	Description
	connectivity is available.

4.5.3 User Requirements

User requirements describe the expectations of the different categories of users who interact with the Parka system. Since the system serves drivers, parking attendants, parking facility operators, and system administrators, each user category had specific needs that guided the design and implementation of the system.

Drivers required a system that could allow them to search for parking facilities, view real-time availability, reserve spaces before arrival, manage vehicles, access reservation codes or QR tokens, make payments, view receipts, and receive notifications.

Parking attendants required a system that could help them verify reservations, scan QR codes, check vehicles in, check vehicles out, register walk-in vehicles, view assigned facility occupancy, and confirm payment or session status.

Parking facility operators required a dashboard for creating facilities, managing zones and spaces, assigning attendants, setting pricing rules, monitoring occupancy, viewing reservations, and tracking revenue records.

System administrators required tools for managing users, verifying operators, approving facilities, suspending accounts, monitoring system activity, and reviewing audit logs.

Table 4.5: User Requirements of the Proposed System

User Category	User Requirements
Driver	Registration, login, vehicle management, parking search, availability viewing, reservation, QR access, payment confirmation, receipt viewing, and notifications.
Parking Attendant	Reservation verification, QR scanning, vehicle check-in, vehicle check-out,

User Category	User Requirements
	walk-in vehicle registration, occupancy viewing, and session status confirmation.
Parking Facility Operator	Facility management, zone management, space management, attendant assignment, pricing management, reservation monitoring, occupancy tracking, and payment reporting.
System Administrator	User management, operator verification, facility approval, account suspension, activity monitoring, and audit log review.

4.6 System Users and Roles

Parka was designed to support four major categories of users: drivers, parking attendants, parking facility operators, and system administrators. Each role was given specific permissions and functions based on its responsibilities in the parking management process.

4.6.1 Driver

The driver is the primary user of the system. The driver uses Parka to search for parking facilities, view available parking spaces, reserve a space, manage vehicle information, present a reservation code during check-in, monitor active sessions, make simulated payments, receive receipts, and provide feedback on parking facilities.

The driver journey begins when a user logs into the system and searches for a nearby or preferred parking facility. The driver then selects a facility, chooses a vehicle, reserves a space, and receives a reservation code or QR token. Upon arrival, the code is verified by the attendant, and the parking session begins.

4.6.2 Parking Attendant

The parking attendant manages vehicle entry and exit at an assigned facility. The attendant uses the system to verify driver reservations, scan QR codes, check vehicles in, check vehicles out, register walk-in vehicles, and confirm session or payment status.

The attendant role is important because it connects the digital reservation system with the physical parking environment. The attendant ensures that the correct vehicle is allowed entry and that the session is properly opened during entry and closed during exit.

4.6.3 Parking Facility Operator

The operator manages one or more parking facilities. The operator uses Parka to add facility details, create parking zones, define individual parking spaces, assign attendants, set pricing rules, monitor occupancy, view reservations, and analyze payments or revenue records.

This role supports facility-level management and gives parking business owners better visibility into their operations.

4.6.4 System Administrator

The system administrator oversees the entire platform. The administrator manages users, verifies operators, approves or suspends parking facilities, monitors system activity, and reviews audit logs. This role ensures that the platform operates securely and that only valid operators and facilities are allowed to participate.

The Parka system documentation confirms that the system was organized around four major user journeys: driver, attendant, operator, and administrator.

4.7 System Design

System design describes how the proposed solution was structured before and during implementation. It presents the logical and physical arrangement of system components, user interactions, data flow, and database relationships.

The design of Parka focused on ensuring that drivers could find and reserve parking spaces easily, attendants could verify parking sessions efficiently, operators could manage facilities effectively, and administrators could monitor the platform securely.

The major design components included use case design, context design, data flow design, reservation workflow, check-in and check-out workflow, system architecture, entity relationship design, and database schema design. The figures were arranged from logical design to physical design so that the chapter first explains how users interact with the system before presenting the technical and database structure.

4.7.1 Use Case Design

The use case design describes the interaction between Parka and its users. The major actors in the system are the driver, parking attendant, parking facility operator, and system administrator.

The driver interacts with the system by creating an account, logging in, adding vehicle details, searching for parking facilities, viewing availability, reserving a space, checking active sessions, making payment, receiving a receipt, and reviewing a facility.

The attendant interacts with the system by logging in, viewing the assigned facility, scanning or verifying reservation codes, checking vehicles in, checking vehicles out, registering walk-in vehicles, and confirming payment status.

The operator interacts with the system by managing parking facilities, creating zones, creating parking spaces, assigning attendants, setting pricing rules, monitoring occupancy, and viewing revenue records.

The administrator interacts with the system by managing users, verifying operators, approving or suspending facilities, and monitoring system activity.

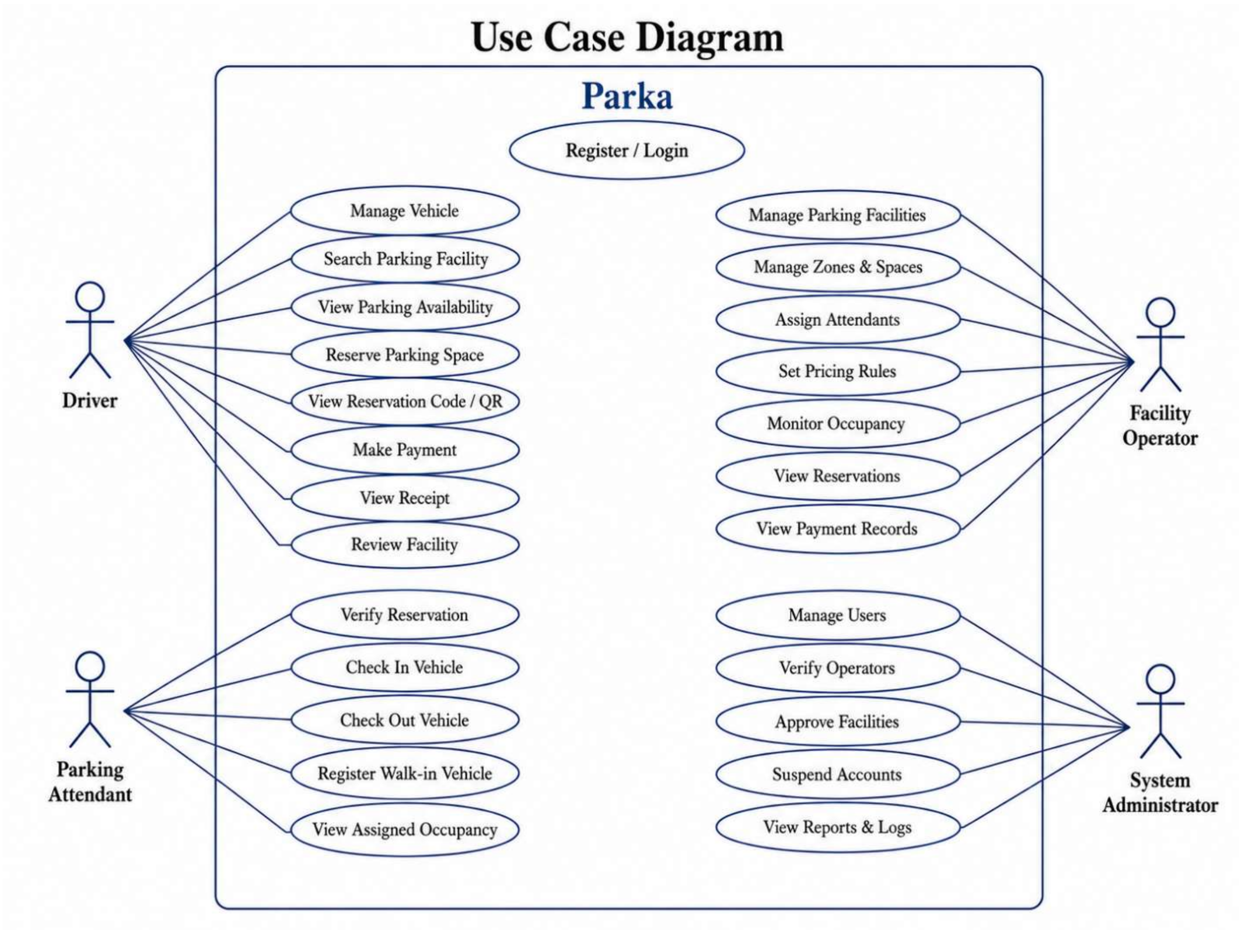
Table 4.6: Data Dictionary for Use Case Processes

Process ID	**Process**	**Description**
1.1	Register / Login	Allows drivers, attendants, operators, and administrators to access the system securely.
1.2	Manage Vehicle	Allows drivers to add, update, and view vehicle details used during parking reservation.
1.3	Search Parking	Allows drivers to search for parking facilities.

Process ID	Process	Description
	Facility	
1.4	View Parking Availability	Allows drivers to view available, reserved, and occupied parking spaces.
1.5	Reserve Parking Space	Allows drivers to book parking spaces before arrival.
1.6	View Reservation Code / QR	Allows drivers to access the reservation code or QR token used during check-in.
1.7	Make Payment	Allows drivers to confirm simulated payment after checkout.
1.8	View Receipt	Allows drivers to view proof of payment after successful payment.
1.9	Verify Reservation	Allows parking attendants to confirm whether a reservation is valid.
1.10	Check In Vehicle	Allows attendants to start a parking session after reservation verification.
1.11	Check Out Vehicle	Allows attendants to close an active parking session when a driver exits.
1.12	Register Walk-in Vehicle	Allows attendants to register vehicles that arrive without prior reservation.
1.13	Manage Parking Facilities	Allows operators to create and manage parking facility records.

Process ID	Process	Description
1.14	Manage Zones and Spaces	Allows operators to define zones and individual parking spaces.
1.15	Assign Attendants	Allows operators to assign attendants to parking facilities.
1.16	Set Pricing Rules	Allows operators to define parking prices.
1.17	Manage Users and Facilities	Allows administrators to control users, verify operators, approve facilities, and monitor activity.

Figure 2:Use Case Diagram of Parka



4.7.2 Context Diagram / Level 0 DFD

The context diagram presents Parka as a single system and shows how it interacts with external users. The main external entities are the driver, parking attendant, parking facility operator, and system administrator.

Drivers send requests such as registration, parking search, reservation, payment, and feedback. The system responds with parking availability, reservation confirmation, QR code, active session information, payment status, and receipts.

Attendants send check-in, check-out, and walk-in vehicle registration requests. The system responds with reservation validation, session status, fee details, and payment confirmation.

Operators send facility, zone, space, attendant, and pricing management requests. The system responds with facility records, occupancy information, reservation records, and revenue reports.

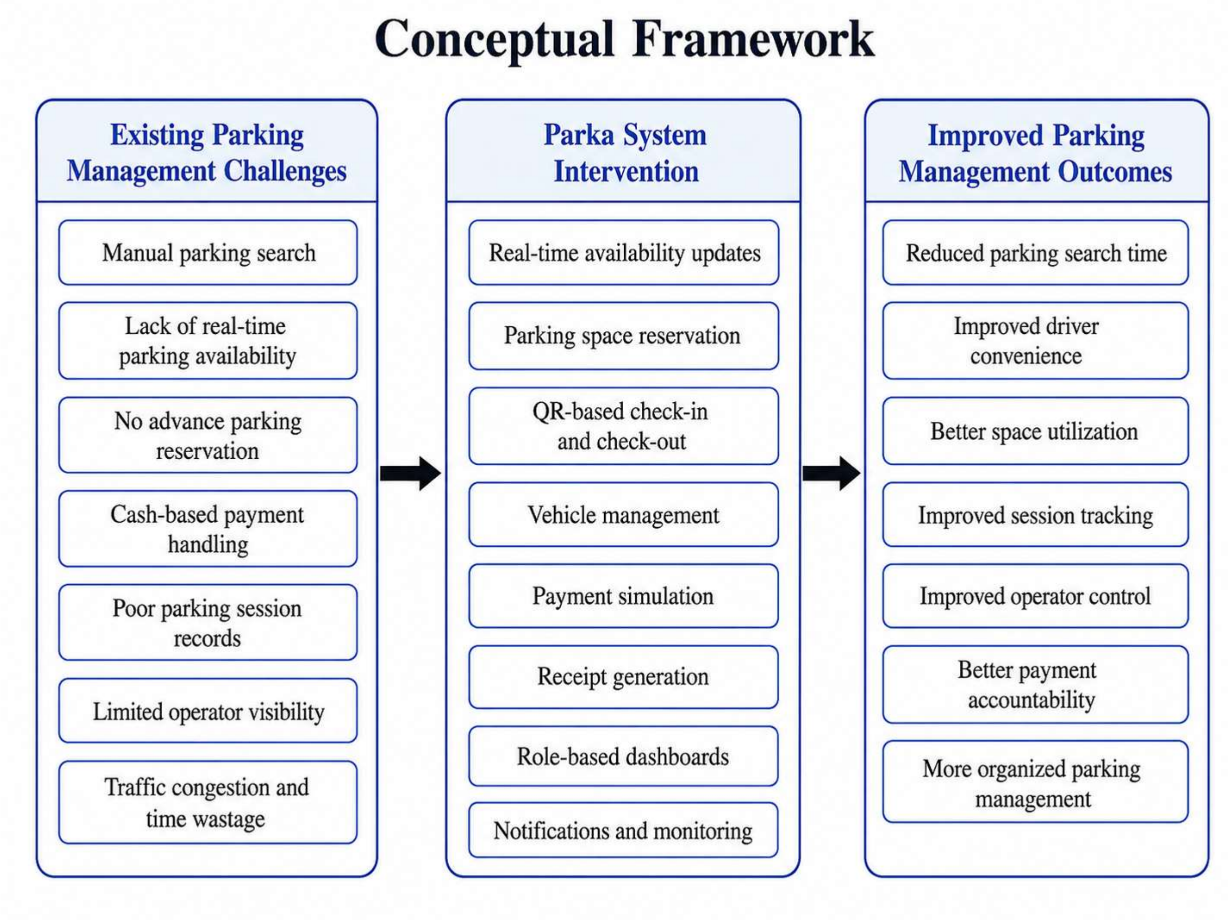
Administrators send verification, approval, monitoring, and user management requests. The system responds with user records, operator details, facility status, and audit information.

Table 4.7: Data Dictionary for Context Diagram Processes

Process ID	**Process**	**Description**
2.1	Driver Data Submission	The driver sends registration details, parking search requests, reservation requests, payment confirmation, and feedback to the system.
2.2	Driver Response Output	The system sends parking availability, reservation confirmation, QR code, receipts, and notifications to the driver.
2.3	Attendant Data Submission	The attendant sends reservation verification, check-in, check-out, and walk-in vehicle details to the system.
2.4	Attendant Response	The system sends reservation status, session details, fee details,

Process ID	**Process**	**Description**
	Output	and payment status to the attendant.
2.5	Operator Data Submission	The operator sends facility details, zones, spaces, attendant assignments, and pricing rules to the system.
2.6	Operator Response Output	The system sends occupancy reports, reservation records, payment records, and revenue reports to the operator.
2.7	Administrator Data Submission	The administrator sends user management, operator verification, facility approval, and monitoring requests to the system.
2.8	Administrator Response Output	The system sends user records, operator status, facility status, and audit logs to the administrator.

Figure 3:Context Diagram / Level 0 DFD of Parka



4.7.3 Data Flow Design

The data flow design shows how information moves through the system. The main data flows include user authentication, parking facility search, reservation processing, QR verification, payment simulation, receipt generation, and real-time notification updates.

When a driver reserves a parking space, the system checks available spaces from the database, assigns a space, creates a reservation record, updates the space status, and returns a reservation code or QR token. When the driver arrives, the attendant verifies the reservation, and the system starts a parking session. During exit, the system calculates the parking fee, processes simulated payment, generates a receipt, and updates the space status to available.

This flow ensures that parking spaces are tracked from booking to completion.

4.7.4 Reservation Workflow

The reservation workflow shows how a driver reserves a parking space using Parka. This workflow was designed to address the absence of advance reservation in the existing parking system.

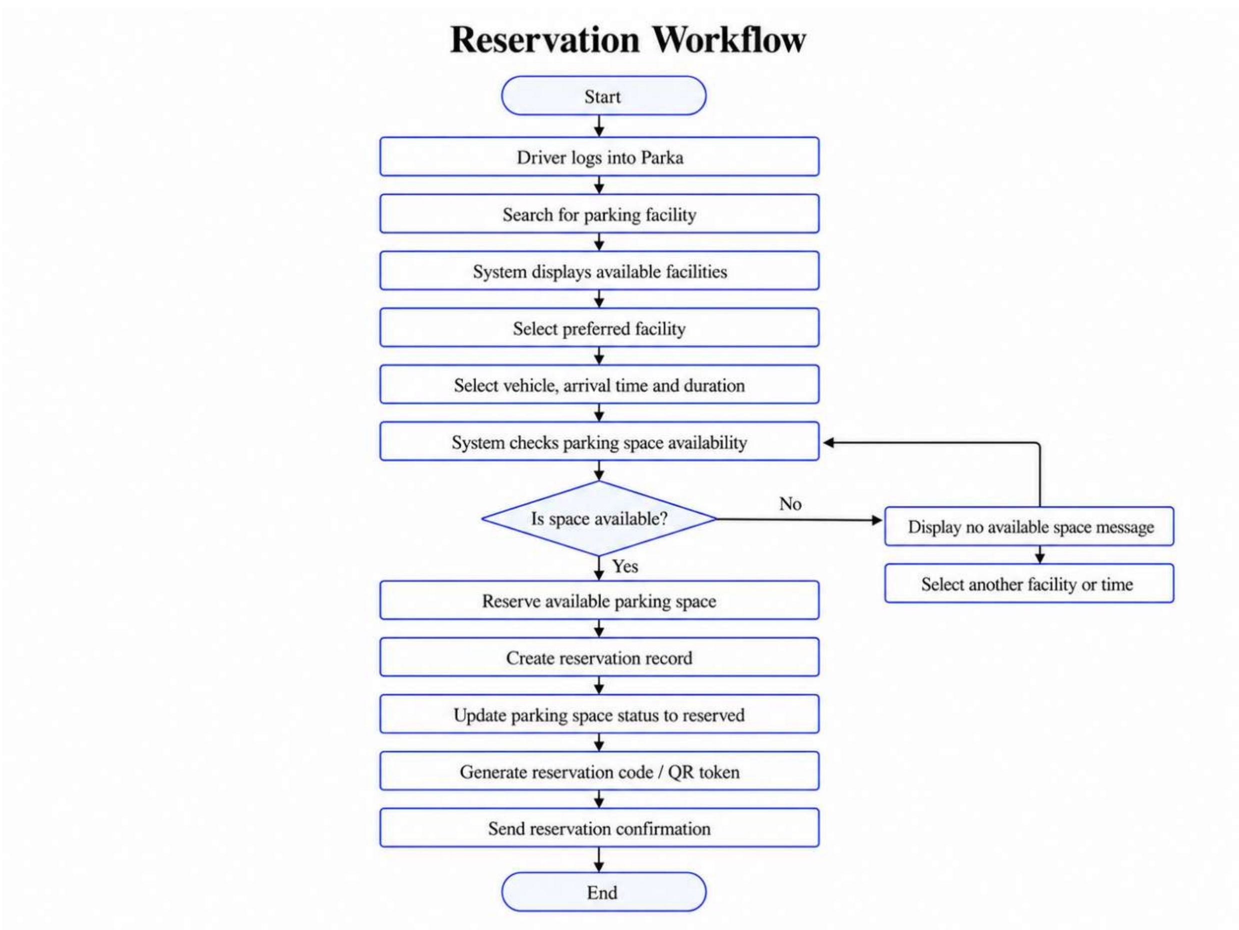
Through this workflow, the driver is able to secure a parking space before arriving at the facility. This reduces uncertainty and saves time that would otherwise be spent searching physically for parking.

Table 4.8: Data Dictionary for Reservation Workflow Processes

Process ID	**Process**	**Description**
3.1	Driver Login	The driver logs into Parka using valid account credentials.
3.2	Parking Facility Search	The driver searches for a preferred or nearby parking facility.
3.3	Display Available Facilities	The system displays facilities and available parking information.
3.4	Facility Selection	The driver selects a preferred parking facility.
3.5	Vehicle and Time Selection	The driver selects a vehicle, arrival time, and expected parking duration.
3.6	Availability Check	The system checks whether parking space is available.
3.7	Space Reservation	If space is available, the system reserves it for the driver.
3.8	Reservation Record Creation	The system creates a reservation record in the database.

Process ID	Process	Description
3.9	Space Status Update	The system updates the parking space status to reserved.
3.10	QR Code Generation	The system generates a reservation code or QR token.
3.11	Reservation Confirmation	The system sends reservation confirmation to the driver.
3.12	No Space Available	If no space is available, the system displays a message and allows the driver to select another facility or time.

Figure 4:Reservation Workflow



4.7.5 Check-in and Check-out Workflow

The check-in and check-out workflow describes how the system manages vehicle entry, parking session tracking, payment, and exit.

This workflow ensures that a parking session is properly recorded from entry to exit. It also allows the system to update parking availability after each check-in and check-out.

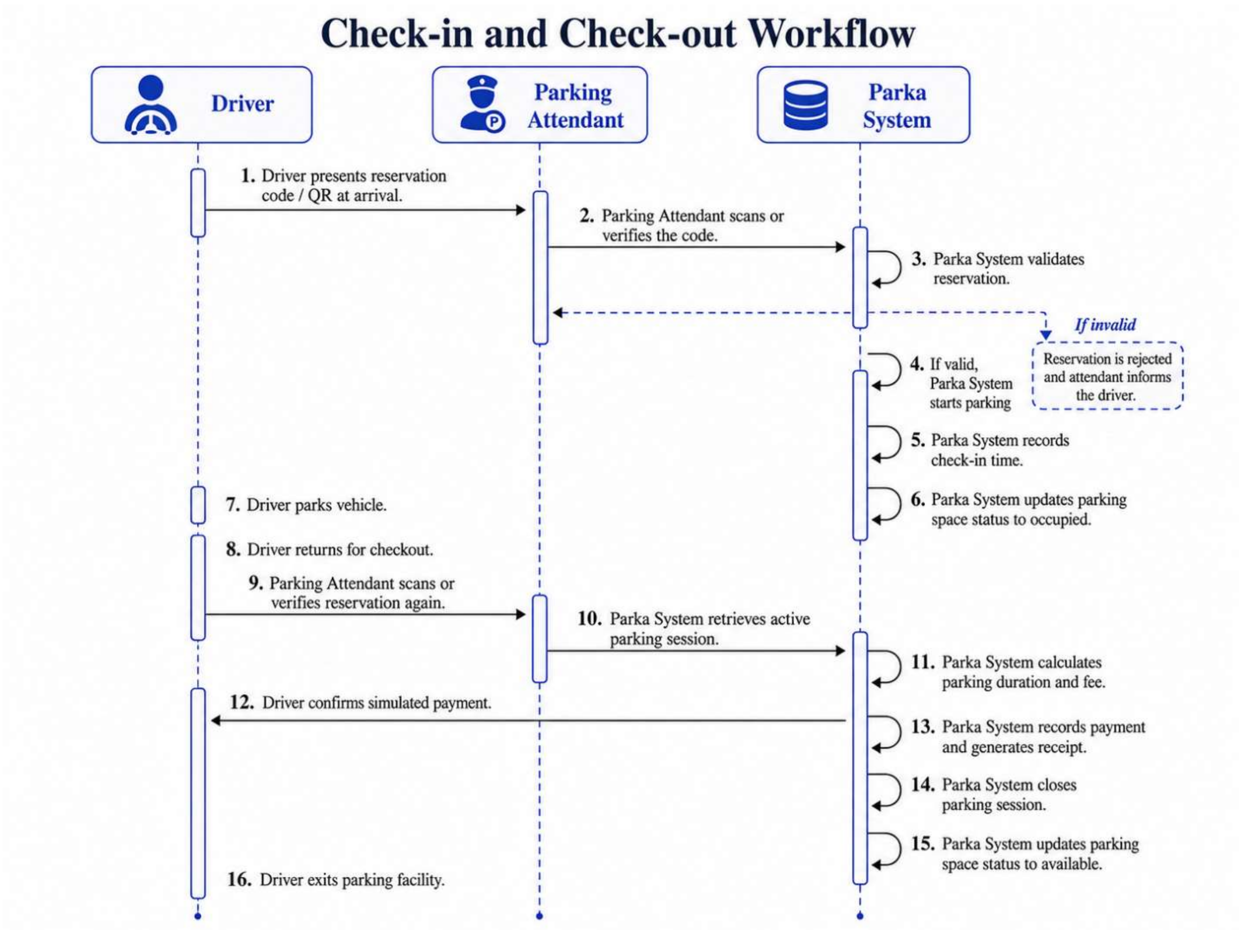
The Parka documentation shows that the implemented operational flow follows this process from booking, check-in, active session tracking, checkout, payment approval, receipt generation, and space availability update.

Table 4.9: Data Dictionary for Check-in and Check-out Processes

Process ID	**Process**	**Description**
4.1	Driver Arrival	The driver arrives at the parking facility and presents the reservation code or QR token.
4.2	Reservation Verification	The parking attendant scans or verifies the code presented by the driver.
4.3	Reservation Validation	The system checks whether the reservation is valid.
4.4	Session Start	If the reservation is valid, the system starts the parking session.
4.5	Check-in Time Recording	The system records the time when the vehicle enters the facility.
4.6	Occupancy Update	The system updates the parking space status to occupied.
4.7	Driver Parking	The driver parks the vehicle in the assigned or available space.

Process ID	Process	Description
4.8	Checkout Request	The driver returns and requests checkout.
4.9	Active Session Retrieval	The system retrieves the active parking session.
4.10	Fee Calculation	The system calculates the parking duration and fee.
4.11	Payment Confirmation	The driver confirms simulated payment.
4.12	Receipt Generation	The system records payment and generates a receipt.
4.13	Session Closure	The system closes the parking session.
4.14	Availability Update	The system updates the parking space status back to available.
4.15	Vehicle Exit	The driver exits the parking facility.

Figure 5:Check-in and Check-out Workflow



4.8 System Architecture

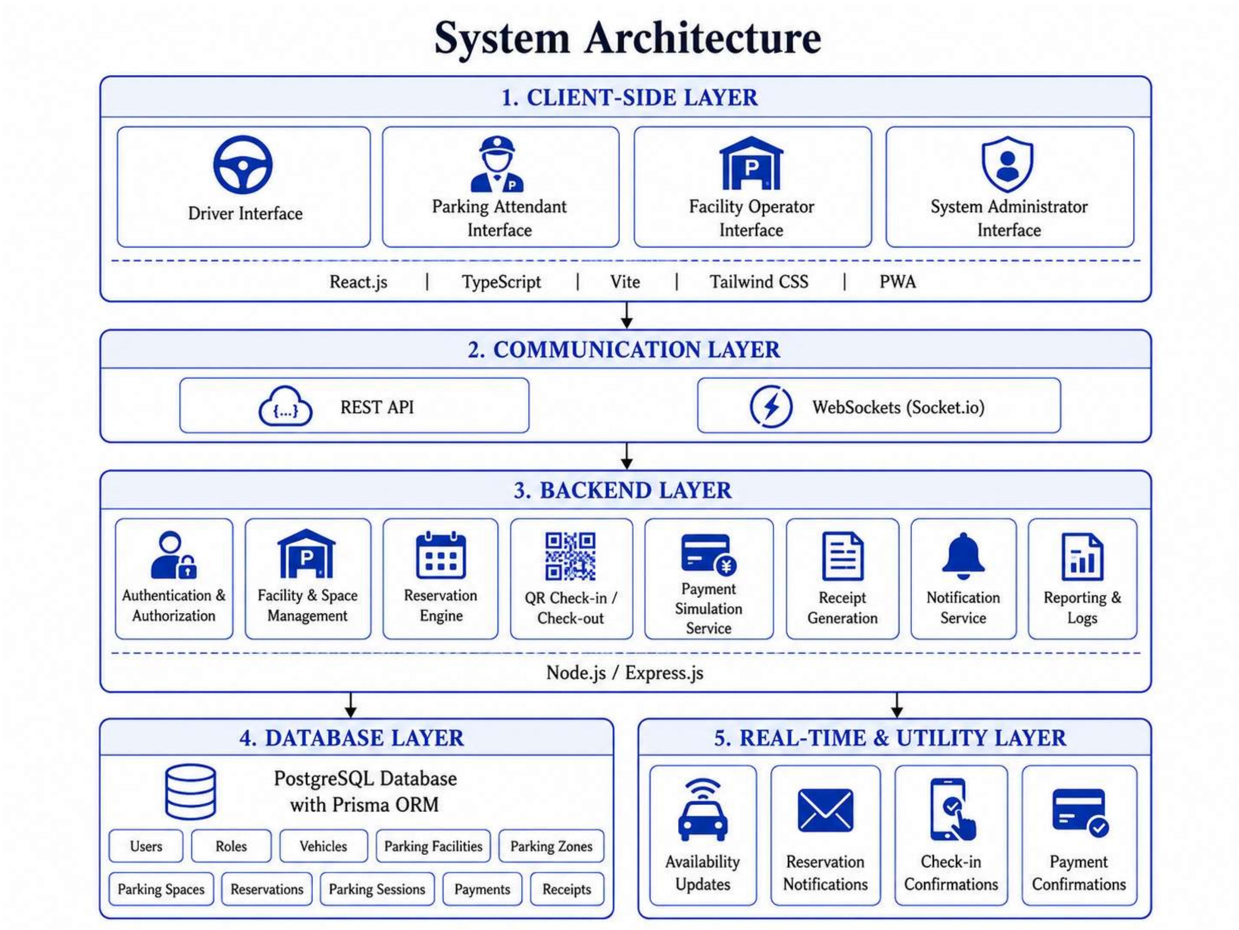
Parka was implemented using a three-tier architecture. This architecture separates the system into the presentation layer, application logic layer, and data layer. The separation of these layers improves scalability, maintainability, security, and system organization.

Table 4.10: Data Dictionary for System Architecture Components

Component ID	**Component**	**Description**
5.1	Client-Side Layer	Provides the user interfaces used by drivers, attendants, operators, and administrators.

Component ID	Component	Description
5.2	Driver Interface	Allows drivers to search, reserve, check sessions, make payments, and view receipts.
5.3	Parking Attendant Interface	Allows attendants to verify reservations and manage check-in and check-out.
5.4	Facility Operator Interface	Allows operators to manage facilities, spaces, attendants, pricing, reservations, and payments.
5.5	System Administrator Interface	Allows administrators to manage users, verify operators, approve facilities, and monitor activity.
5.6	Communication Layer	Supports communication between frontend and backend through REST APIs and WebSockets.
5.7	Backend Layer	Handles authentication, business logic, reservations, QR verification, payment simulation, receipts, and notifications.
5.8	Database Layer	Stores users, vehicles, facilities, spaces, reservations, sessions, payments, receipts, and notifications.
5.9	Real-Time Layer	Supports availability updates, reservation notifications, check-in confirmations, and payment confirmations.

Figure 6:System Architecture



4.8.1 Presentation Layer

The presentation layer is the frontend interface used by drivers, attendants, operators, and administrators. It was implemented as a React-based Single Page Application. The frontend provides user dashboards, forms, maps, reservation pages, QR display pages, scanner interfaces, operator management screens, and administrator monitoring screens.

The frontend was designed to support both mobile and desktop users. This was important because drivers and attendants are likely to use mobile devices, while operators and administrators may prefer larger screens for management dashboards.

4.8.2 Application Logic Layer

The application logic layer was implemented using Node.js and Express. This layer handles the main business logic of the system, including authentication, role-based access control, parking facility management, reservation processing, QR verification, parking session management, payment simulation, receipt generation, and notifications.

The backend exposes RESTful APIs that allow the frontend to communicate with the database and execute system operations securely.

4.8.3 Data Layer

The data layer was implemented using PostgreSQL. The database stores users, roles, profiles, vehicles, parking facilities, parking zones, parking spaces, reservations, parking sessions, payments, receipts, notifications, reviews, and support records.

PostgreSQL was selected because it supports relational data integrity, constraints, indexing, and transactional operations. These features were important in preventing issues such as double-booking and inconsistent parking space status.

4.8.4 Real-Time Communication Layer

Parka also uses Socket.io to support real-time communication. This allows the system to instantly update users when parking availability changes, when a reservation is created, when a vehicle is checked in, when a payment succeeds, or when a notification is generated.

The system documentation states that Socket.io was used to broadcast real-time updates such as facility availability updates and notification creation events.

4.9 Database Design

The database was designed using PostgreSQL and structured to support users, roles, parking facilities, spaces, reservations, sessions, payments, receipts, notifications, reviews, and support records. The database model supports relational integrity through primary keys, foreign keys, constraints, and indexes.

The major database entities include roles, users, driver profiles, operator profiles, attendant profiles, vehicles, parking facilities, parking zones, parking spaces, reservations, parking sessions, payments, receipts, notifications, reviews, and support tickets.

The database was designed to support the complete parking process from user registration to reservation, check-in, payment, receipt generation, and checkout. It also supports operator and administrator management functions.

4.9.1 Entity Relationship Diagram

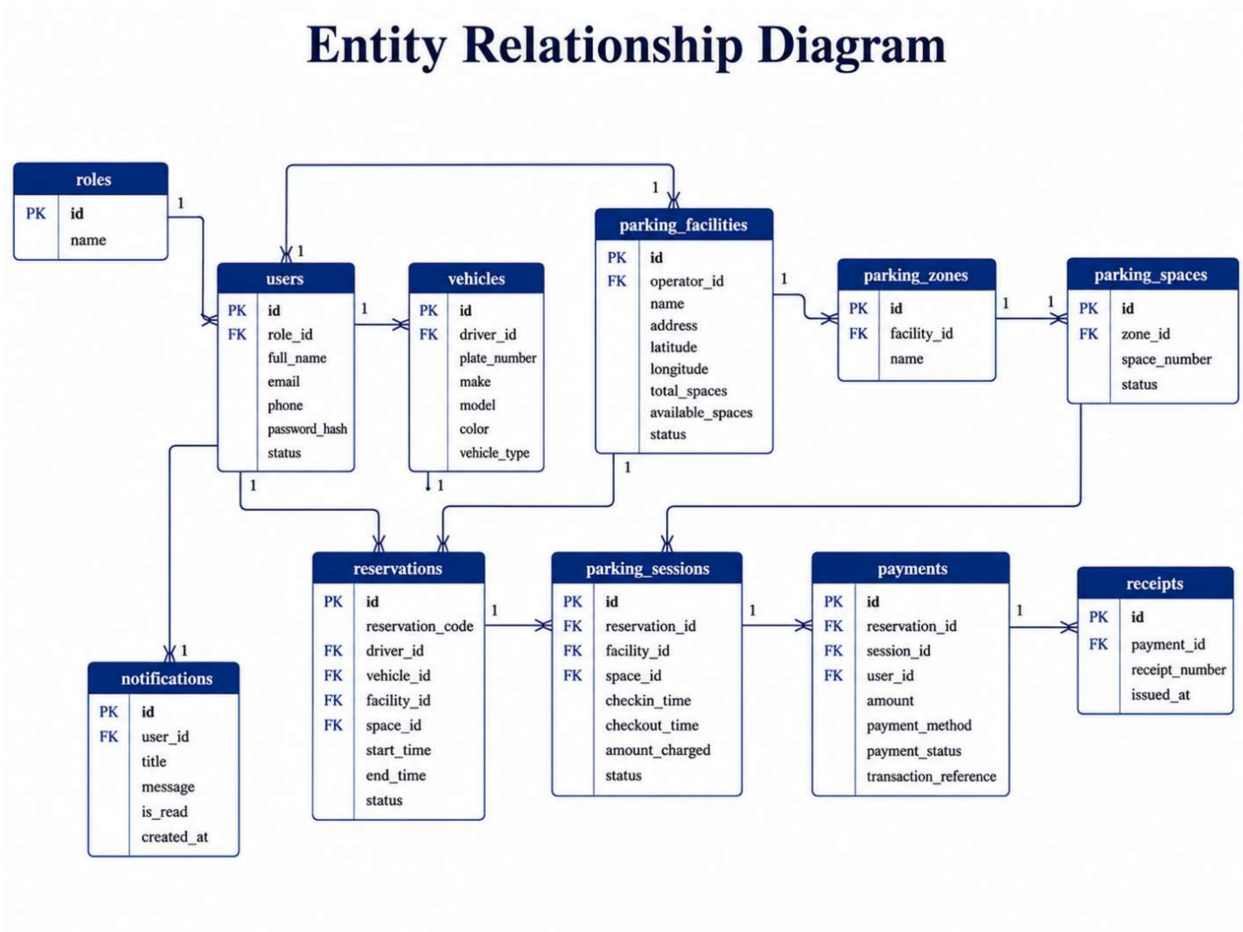
The Entity Relationship Diagram shows the logical relationship between the main entities in the Parka database. It explains how users, roles, vehicles, parking facilities, parking zones, parking spaces, reservations, parking sessions, payments, receipts, and notifications relate to one another.

Table 4.11: Data Dictionary for Database Entities

Entity ID	**Entity**	**Description**
6.1	Roles	Stores role definitions such as driver, attendant, operator, and administrator.
6.2	Users	Stores account details for all system users.
6.3	Vehicles	Stores vehicle information belonging to drivers.
6.4	Parking Facilities	Stores details of parking facilities managed by operators.
6.5	Parking Zones	Stores sections or zones within parking facilities.
6.6	Parking Spaces	Stores individual parking spaces and their statuses.
6.7	Reservations	Stores parking bookings made by drivers.

Entity ID	Entity	Description
6.8	Parking Sessions	Stores active and completed parking sessions.
6.9	Payments	Stores payment transaction records.
6.10	Receipts	Stores receipts generated after successful payment.
6.11	Notifications	Stores system notifications sent to users.

Figure 7: Entity Relationship Diagram



4.9.2 Database Schema

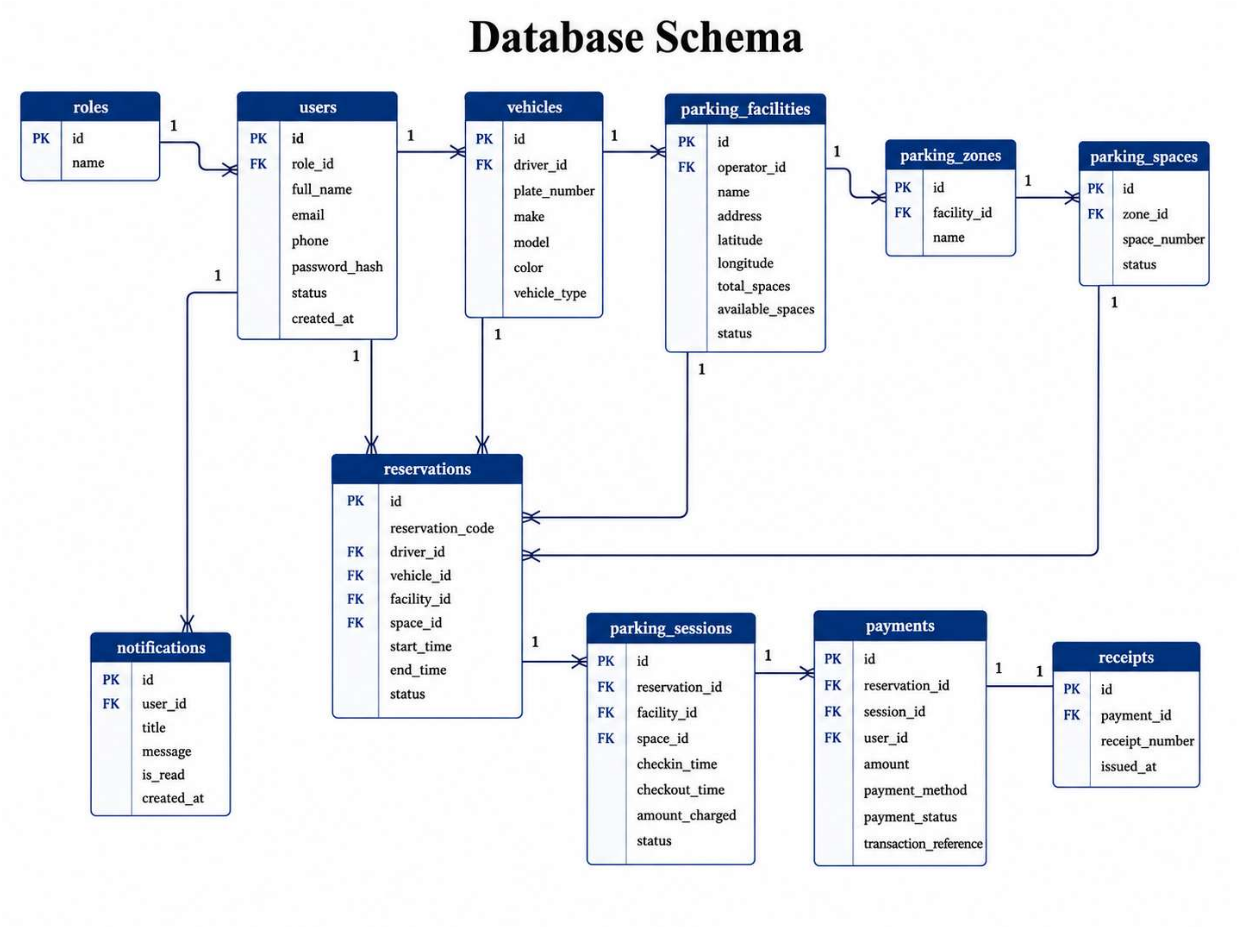
The database schema presents the physical structure of the main database tables, including primary keys, foreign keys, and key attributes. It shows how Parka stores and organizes information needed for user management, vehicle management, facility management, reservations, sessions, payments, receipts, and notifications.

Table 4.12: Data Dictionary for Main Database Tables

Table	Key Fields	Purpose
roles	id, name	Stores user role definitions such as driver, attendant, operator, and administrator.
users	id, role_id, full_name, email, phone, password_hash, status, created_at	Stores user login and account details.
vehicles	id, driver_id, plate_number, make, model, color, vehicle_type	Stores vehicles owned by drivers.
parking_facilities	id, operator_id, name, address, latitude, longitude, total_spaces, available_spaces, status	Stores parking facility information.
parking_zones	id, facility_id, name	Stores facility sections such as floors, blocks, or zones.
parking_spaces	id, zone_id, space_number, status	Stores individual parking spaces.
reservations	id, reservation_code, driver_id, vehicle_id, facility_id, space_id, start_time, end_time, status	Stores parking bookings.

Table	Key Fields	Purpose
parking_sessions	id, reservation_id, facility_id, space_id, checkin_time, checkout_time, amount_charged, status	Stores active and completed parking sessions.
payments	id, reservation_id, session_id, user_id, amount, payment_method, payment_status, transaction_reference	Stores payment transaction records.
receipts	id, payment_id, receipt_number, issued_at	Stores payment receipt records.
notifications	id, user_id, title, message, is_read, created_at	Stores user notifications.

Figure 8: Database Schema



4.9.3 Database Relationships

The database relationships were designed to maintain consistency between users, parking facilities, reservations, sessions, and payments.

A user can have one role, and the role determines the user's permissions in the system. A driver can own several vehicles. A driver can make many reservations, and each reservation belongs to a parking facility and may be linked to a specific parking space. A parking facility belongs to an operator and can have many parking zones. Each parking zone can have many individual parking spaces.

A reservation can lead to a parking session when the driver checks in. A parking session records the time the vehicle entered and exited the facility. A payment may be linked to a reservation or parking session, and a successful payment generates a receipt. Notifications are linked to users and are used to inform them about system events.

These relationships ensure that the system maintains accurate records throughout the parking management process.

4.9.4 Data Integrity and Transaction Safety

Data integrity was important because a parking system must prevent errors such as assigning the same space to more than one driver. Parka was therefore implemented with database constraints, unique identifiers, status fields, and transaction-safe operations.

For example, during reservation and check-in, the system updates the reservation status and parking space status in a controlled process. When a driver checks in, the reservation becomes active, and the parking space changes from reserved to occupied. When the driver checks out and payment is completed, the parking session is closed, and the parking space becomes available again.

The system documentation explains that database transactions were used to prevent double-booking and race conditions during critical operations such as check-in and space status updates.

4.10 System Implementation

System implementation involved transforming the system design into a working application. Parka was implemented as a web-based smart parking management and reservation platform with separate interfaces for drivers, attendants, operators, and administrators.

The implementation was divided into frontend development, backend development, database implementation, authentication and authorization, facility management, reservation processing, QR-based check-in and check-out, payment simulation, receipt generation, real-time updates, and notifications.

4.10.1 Frontend Implementation

The frontend was implemented using React, Vite, TypeScript, Tailwind CSS, and supporting user interface libraries. It provides the visual interface through which users interact with the system.

The frontend includes different dashboards depending on the user role. Drivers access parking search, facility details, reservations, active sessions, payment screens, and receipts. Attendants access scanner and session management interfaces. Operators access facility management, zone management, space management, attendant assignment, pricing, and analytics screens. Administrators access user management, operator verification, facility approval, and system monitoring screens.

The frontend was designed to be responsive so that it can work on both mobile phones and computers. This was important because the system serves different users in different environments. Drivers and attendants may use mobile phones, while operators and administrators may prefer computers for dashboard management.

4.10.2 Backend Implementation

The backend was implemented using Node.js and Express. It handles the application logic and provides API endpoints used by the frontend. The backend manages user authentication, role-based access, facility records, parking spaces, reservations, sessions, payments, receipts, notifications, and system administration.

The backend was organized into modules to improve maintainability. These modules included authentication, facility management, reservation management, QR verification, payment simulation, user management, notifications, and administrative controls.

Security middleware such as CORS, Helmet, rate limiting, and error handling was used to improve API safety and reliability. The backend communicates with the PostgreSQL database using database queries and connection pooling.

The Parka documentation identifies the backend as an Express API with modular namespaces, security middleware, custom error handlers, and database communication through PostgreSQL.

4.10.3 Authentication and Authorization

Authentication was implemented to ensure that only registered users can access protected system functions. Users log in using their credentials, and the system validates them before granting access.

Passwords were hashed before storage to protect user accounts. JSON Web Tokens were used to maintain authenticated sessions and authorize access to protected routes. Role-based access control was used to ensure that each user can only access features allowed for their role.

For example, drivers can search and reserve parking, but they cannot approve facilities. Attendants can scan reservation codes and manage check-in or check-out, but they cannot create operator facilities. Operators can manage their own facilities, zones, spaces, and attendants, while administrators can monitor and control the overall system.

The system documentation states that Parka used bcrypt for password hashing and JWT for API protection.

4.10.4 Facility and Space Management

Facility and space management was implemented to allow parking operators to register and manage parking facilities. An operator can add facility details such as facility name, location, address, total spaces, price per hour, and operational details.

The operator can also divide a facility into parking zones. A zone may represent a floor, section, or block within a facility. Each zone contains individual parking spaces. Each parking space has a status such as available, reserved, occupied, or unavailable.

This structure allows the system to track availability accurately and support reservations at the facility and space level.

4.10.5 Reservation Engine

The reservation engine is one of the core components of Parka. It allows drivers to book parking spaces before arriving at a facility. During reservation, the system checks whether a selected facility has available spaces. If a space is available, the system assigns or reserves the space and creates a reservation record.

After a successful reservation, the system generates a reservation code or QR token. The driver presents this code at the parking facility for verification by the attendant.

The reservation engine also updates parking availability so that other drivers can see the correct number of available spaces. This reduces the chance of double-booking and improves reliability.

4.10.6 QR-Based Check-in and Check-out

QR-based check-in and check-out were implemented to connect the digital reservation process with physical parking operations. When a driver arrives at a parking facility, the attendant scans or verifies the reservation QR code. The system validates the reservation and starts a parking session.

When check-in is completed, the parking space status changes from reserved to occupied. The active session records the check-in time and vehicle details.

At checkout, the attendant scans or verifies the reservation again. The system retrieves the active parking session, calculates the duration, determines the fee, and prepares the payment process. After payment is completed, the system marks the session as completed and changes the parking space status back to available.

4.10.7 Parking Fee Calculation

Parking fee calculation was implemented based on the duration of the parking session and the pricing rules defined for the facility. The system calculates the time difference between check-in and checkout. The duration is then converted into billable units, and the price is calculated using the facility's rate.

For example, if a facility charges per hour, the system calculates the number of hours parked and multiplies it by the hourly rate. Where partial hours are involved, the system may round up based on the pricing logic used.

The Parka documentation explains that parking fees are calculated dynamically at checkout by extracting the duration, rounding minutes into billable hours, and applying the hourly rate or pricing rules.

4.10.8 Payment Simulation and Receipt Generation

Payment was implemented as a simulator to represent mobile money-style parking payments. This approach was used because direct integration with live MTN Mobile Money or Airtel Money APIs was outside the scope of the academic project.

When a driver checks out, the system displays the calculated bill. The driver then confirms payment using the simulated payment process. After successful payment, the system records the transaction, marks the payment as completed, generates a receipt, closes the parking session, and updates the parking space as available.

Receipt generation provides proof of payment and improves accountability for both drivers and parking operators.

4.10.9 Real-Time Updates and Notifications

Real-time updates were implemented using Socket.io. This allowed the system to instantly notify users when important events occurred. For example, when a driver reserves a parking space, facility availability is updated. When a vehicle checks in, the space status changes to occupied. When a vehicle checks out and payment is completed, the space becomes available again.

Notifications were used to inform users about reservations, check-in confirmation, payment success, and session updates. Real-time communication improves user experience because users do not need to manually refresh the system to see updated information.

The system documentation states that Parka uses real-time channels to push events such as notification creation and facility availability updates.

4.10.10 Administrator Implementation

The administrator module was implemented to support system-wide control and monitoring. The administrator can manage users, verify or suspend operators, approve facilities, and monitor system activity.

This module ensures that the platform remains controlled and that only approved facilities and operators are allowed to serve drivers. It also supports accountability through monitoring and audit functions.

4.11 System Interfaces

System interfaces are the visible screens through which users interact with Parka. The interfaces were designed to support the main functions of each user role. Since Parka serves drivers, attendants, operators, and administrators, each interface was designed according to the needs of its users.

4.11.1 Login and Registration Interface

The login and registration interface allows users to create accounts and access the system securely. The interface collects user details such as name, email, phone number, password, and role. After login, the user is redirected to the dashboard that matches their role.

4.11.2 Driver Interface

The driver interface allows users to search for parking facilities, view availability, make reservations, manage vehicles, monitor active sessions, make simulated payments, and view receipts. The interface was designed to be mobile-friendly because drivers are likely to use the system while moving.

4.11.3 Attendant Interface

The attendant interface allows parking attendants to verify reservations, scan QR codes, check vehicles in, check vehicles out, register walk-in vehicles, and view facility occupancy. The interface was designed for quick use at entry and exit points.

4.11.4 Operator Interface

The operator interface allows parking facility owners or managers to add facilities, manage parking zones, create spaces, assign attendants, define pricing rules, and monitor occupancy or revenue. The operator dashboard provides management tools that improve control over parking operations.

4.11.5 Administrator Interface

The administrator interface allows the system administrator to manage users, verify operators, approve parking facilities, suspend accounts, and monitor audit logs. This interface supports platform governance and ensures that the system remains reliable and secure.

4.12 Security Implementation

Security was considered during implementation because the system handles user accounts, vehicle details, reservation records, and payment-related information. The security mechanisms implemented included password hashing, token-based authentication, role-based access control, secure API middleware, and protected routes.

Passwords were hashed before storage so that plain-text passwords were not kept in the database. JWT authentication was used to allow secure access to protected APIs. Role-based access control ensured that users could only access functions assigned to their roles. For example, a driver could not access operator management functions, and an attendant could not approve facilities.

Security middleware such as Helmet, CORS, and rate limiting was also used to reduce common web application risks. The system documentation identifies hashed passwords, JWT protection, and rate limiting as part of Parka's security protocols.

4.13 Deployment

The system was prepared for deployment as a web-based application. The frontend was built as a React Single Page Application, while the backend was deployed as an Express server connected to a PostgreSQL database. The system was designed to support deployment on cloud platforms or server environments.

The deployment structure followed a separation between frontend, backend, and database services. This allowed easier maintenance, independent updates, and future scalability.

The system documentation identifies Dockerized services and Render cloud deployment as part of the scalable deployment model prepared for Parka.

4.14 Chapter Summary

This chapter presented the system analysis, design, and implementation of Parka. It began by analyzing the existing parking management system in Kampala and identifying its strengths and weaknesses. The strengths included simplicity, low initial technology cost, human assistance from attendants, flexibility in handling parking situations, and an existing payment culture. The weaknesses included lack of real-time availability, absence of advance reservation, manual record keeping, cash-based payments, poor session tracking, and poor operator visibility.

The chapter then presented the requirements of the proposed system, including functional, non-functional, and user requirements. It also described the major system users and their roles: drivers, parking attendants, parking facility operators, and system administrators.

The system design section arranged the diagrams from logical design to physical design. The logical design included the use case diagram, context diagram, reservation workflow, and check-in and check-out workflow. The physical design included the system architecture, entity relationship diagram, and database schema. Data dictionaries were provided before each figure to explain the major processes, components, and data elements represented in the diagrams.

Overall, Parka was successfully designed and implemented as a smart parking management and reservation system that addresses the limitations of manual parking systems by enabling drivers to search, reserve, check in, pay, and receive receipts digitally, while also allowing operators and administrators to manage parking operations more efficiently.

CHAPTER FIVE: RESULTS, SYSTEM EVALUATION, DISCUSSION, SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the results, system evaluation, discussion, summary, conclusion, recommendations, and areas for future work based on the design and implementation of Parka, a real-time parking management and space reservation system for Kampala. The chapter focuses on the findings obtained from the study, the results produced by the implemented system, and the extent to which the system addressed the parking management challenges identified in Kampala.

The chapter also evaluates the developed system in terms of functionality, usability, performance, security, reliability, and data integrity. The discussion is organized according to the study objectives, which were to review the current parking management system, design a suitable system, implement the system, and evaluate its effectiveness in improving parking management.

Parka was developed as a web-based real-time parking management and space reservation system that enables drivers to search for parking facilities, view availability, reserve spaces, check in and check out using QR-based verification, make simulated payments, and receive receipts. The system also enables parking attendants, operators, and administrators to manage parking operations digitally.

5.2 Presentation of Research Findings

The research findings confirmed that the existing parking management practices in Kampala were largely manual and inefficient. The findings showed that drivers experienced challenges in locating available parking spaces, parking attendants relied on manual processes, and parking operators lacked digital tools for monitoring facility occupancy and revenue.

The findings are presented under the following areas: existing parking management practices, challenges faced by drivers, challenges faced by parking attendants, challenges faced by parking operators, and user readiness to use a digital parking system.

5.2.1 Findings on Existing Parking Management Practices

The study found that parking management in many parts of Kampala still depends heavily on manual methods. Drivers usually locate parking spaces by physically moving from one place to another or by asking parking attendants. Parking availability is not communicated in real time, and most parking facilities do not provide advance space reservation.

The existing parking process is therefore reactive rather than planned. Drivers only know whether a parking space is available after reaching the location. This creates uncertainty and contributes to delays, especially in busy urban areas.

The study also found that parking attendants manually monitor vehicle entry, parking duration, and exit. In some cases, parking records are written on paper or remembered by the attendant. Such methods are vulnerable to errors, poor accountability, and limited reporting.

5.2.2 Findings on Challenges Faced by Drivers

The study found that drivers faced several challenges in the existing parking system. The major challenge was the difficulty of finding available parking spaces, especially in high-demand areas. Drivers often moved from one parking area to another before securing a space.

Another challenge was the lack of prior information. Since drivers could not check parking availability before arrival, they wasted time and fuel searching physically. This also increased frustration and contributed to traffic congestion.

The study also found that drivers lacked a reliable way of reserving parking spaces in advance. This made parking uncertain, especially for users travelling to busy commercial areas, offices, hospitals, shopping areas, and other urban destinations.

Table 5.1: Challenges Faced by Drivers

Challenge	Effect on Drivers
Lack of real-time parking	Drivers could not know where spaces were available before

Challenge	Effect on Drivers
information	arrival
Physical searching for spaces	Increased time wastage and fuel consumption
No advance reservation	Drivers could not secure parking before reaching a destination
Manual payment process	Reduced convenience and limited receipt availability
Uncertainty of parking availability	Increased frustration and poor user experience

5.2.3 Findings on Challenges Faced by Parking Attendants

The study found that parking attendants experienced operational challenges because most parking processes were handled manually. Attendants had to identify available spaces physically, guide vehicles, collect payments, and monitor entry and exit without a centralized digital system.

Manual handling made it difficult to track active parking sessions accurately. In cases where several vehicles were entering and exiting a facility, attendants could easily make mistakes in recording parking duration, fee calculation, and payment confirmation.

The study also found that attendants lacked digital tools for scanning reservations or verifying pre-booked spaces. This meant that the parking process depended on physical presence and verbal communication rather than system-supported verification.

Table 5.2: Challenges Faced by Parking Attendants

Challenge	Effect on Parking Operations
Manual vehicle tracking	Increased risk of errors in session records
No QR or digital verification	Difficult to confirm reservations efficiently
Manual fee handling	Increased chances of incorrect charges
No real-time dashboard	Attendants could not easily monitor facility occupancy
Paper-based or informal records	Reduced accountability and reporting accuracy

5.2.4 Findings on Challenges Faced by Parking Operators

The study found that parking facility operators lacked centralized tools for managing facilities, spaces, attendants, reservations, and payments. Most operators depended on attendants to report occupancy and revenue information manually.

This made it difficult for operators to know the exact number of available, occupied, or reserved spaces at a given time. It also limited their ability to analyze revenue, identify peak parking periods, monitor attendant activity, and make data-driven decisions.

The findings showed that operators needed a system that could provide facility management, space tracking, reservation records, payment records, and occupancy monitoring.

Table 5.3: Challenges Faced by Parking Operators

Challenge	**Effect on Operators**
Lack of centralized facility dashboard	Operators could not monitor parking operations effectively
Manual revenue tracking	Increased risk of revenue leakage and poor accountability
Poor space utilization data	Difficult to identify underused or overcrowded areas
Limited reservation records	Operators could not plan space usage in advance
No digital reports	Reduced ability to analyze performance

5.2.5 Findings on User Readiness to Use a Digital Parking System

The study found that a digital parking system would be useful to drivers, attendants, and operators. Drivers needed a platform that could show parking availability before arrival and allow advance reservation. Attendants needed a system that could support check-in and check-out verification. Operators needed a dashboard for managing facilities, spaces, sessions, attendants, and revenue.

The findings therefore supported the need for Parka as a digital system for real-time parking management and space reservation.

5.3 System Results

This section presents the results of the developed Parka system. The system was successfully implemented with four major user portals: driver portal, attendant portal, operator portal, and administrator portal. Each portal was designed to support the specific activities of its user category.

The system also produced key results in real-time parking availability, reservation, QR-based check-in and check-out, payment simulation, receipt generation, notifications, and parking facility management.

5.3.1 Overview of the Developed Parka System

The developed system provides a digital parking management environment where drivers, attendants, operators, and administrators interact through role-based interfaces. The system allows parking facilities to be registered, spaces to be managed, drivers to reserve parking, attendants to verify parking sessions, and operators to monitor facility performance.

Parka was implemented as a web-based system using a React frontend, Node.js and Express backend, PostgreSQL database, and Socket.io for real-time updates. The system documentation shows that Parka was structured as a digital parking marketplace connecting drivers, parking facility operators, gate attendants, and administrators in a single real-time ecosystem.

5.3.2 Driver Portal Results

The driver portal was successfully implemented to allow drivers to search for parking facilities, view availability, make reservations, manage vehicles, monitor active sessions, complete simulated payments, and view receipts.

Through the driver portal, a user can log in, search for available parking facilities, choose a facility, select a vehicle, specify parking details, and reserve a space. After reservation, the system generates a reservation code or QR token that is used during check-in.

The driver portal also allows the driver to view an active parking session after check-in. At checkout, the driver receives the calculated parking charge and completes payment through the payment simulator.

5.3.3 Attendant Portal Results

The attendant portal was successfully implemented to support parking facility entry and exit operations. The attendant can log in, view the assigned facility, scan or verify reservation codes, check in vehicles, check out vehicles, and register walk-in vehicles.

The QR-based check-in function allows the attendant to verify whether a driver has a valid reservation. Once verified, the system starts a parking session and changes the space status to occupied. During checkout, the attendant verifies the vehicle again, and the system calculates the parking fee based on the session duration.

This result improved the manual parking process by introducing digital verification and session tracking.

5.3.4 Operator Portal Results

The operator portal was successfully implemented to allow parking facility owners or managers to control their parking operations. Through the operator portal, operators can create parking facilities, configure zones, define spaces, assign attendants, set pricing rules, and monitor occupancy.

The operator portal also provides records of reservations, parking sessions, payments, and facility performance. This gives operators better visibility into facility usage and revenue.

The system documentation indicates that the operator journey includes facility portfolio management, zone and space mapping, pricing configuration, analytics, and transaction records.

5.3.5 Administrator Portal Results

The administrator portal was successfully implemented to support platform-level management. The administrator can manage users, verify operators, approve facilities, suspend accounts, and monitor system activity.

This portal ensures that the system remains controlled and secure. It also helps prevent unauthorized operators or invalid facilities from operating on the platform.

5.3.6 Real-Time Parking Availability Results

The system successfully supported real-time updates of parking availability. When a driver reserved a parking space, the availability count was updated. When a vehicle checked in, the space status changed to occupied. When a vehicle checked out and payment was completed, the space status changed back to available.

Socket.io was used to send real-time updates across the system. This allowed users to receive updated information without manually refreshing the page. The Parka documentation states that the system used real-time synchronization to broadcast parking space occupancy, check-in confirmations, and payment success updates.

5.3.7 Reservation and QR Check-in/Check-out Results

The reservation feature was successfully implemented. Drivers were able to reserve parking spaces and receive reservation codes or QR tokens. The attendant could verify the reservation during arrival and start the parking session.

The check-in and check-out workflow was also successfully implemented. During check-in, the system validated the reservation, started a parking session, and updated the parking space status. During checkout, the system calculated the duration and parking fee, then completed the session after payment.

The Parka system documentation describes the operational flow from booking to check-in, active session tracking, checkout, payment, and receipt generation.

5.3.8 Payment Simulation and Receipt Generation Results

The system successfully implemented payment simulation and receipt generation. Since live mobile money integration was outside the project scope, the system used a simulated MTN/Airtel Money-style payment process.

At checkout, the system calculated the parking fee and displayed the bill to the driver. After the driver confirmed payment, the system marked the payment as successful and generated a receipt. The receipt provided proof of payment and improved accountability.

The Parka documentation explains that the payment simulator was designed to represent MTN and Airtel Money-style transactions and that receipts were generated after successful payment.

5.4 System Evaluation

System evaluation was conducted to determine whether Parka met its intended objectives and requirements. The system was evaluated in terms of functionality, usability, performance, security, reliability, and data integrity.

5.4.1 Functional Evaluation

Functional evaluation focused on whether the system features worked as expected. The major functions tested included user registration, login, facility search, reservation, QR check-in, QR check-out, fee calculation, payment simulation, receipt generation, operator facility management, and administrator monitoring.

Table 5.4: Functional Evaluation Results

Function Tested	Expected Result	Actual Result	Status
User registration	User account should be created	Account was created successfully	Passed
User login	User should access correct dashboard	User accessed role-based dashboard	Passed

Function Tested	Expected Result	Actual Result	Status
Facility search	Driver should view parking facilities	Facilities were displayed successfully	Passed
Availability display	Driver should view available spaces	Availability was displayed	Passed
Parking reservation	Driver should reserve available space	Reservation was created successfully	Passed
QR generation	System should generate reservation code or QR	Reservation code/QR was generated	Passed
QR check-in	Attendant should verify reservation and start session	Session started successfully	Passed
QR check-out	Attendant should close active session	Session was closed successfully	Passed
Fee calculation	System should calculate charges	Charges were calculated	Passed
Payment simulation	Payment should be processed	Payment was marked successful	Passed
Receipt generation	Receipt should be generated after payment	Receipt was generated	Passed
Operator facility management	Operator should manage facilities	Facility management worked successfully	Passed
Admin monitoring	Administrator should manage users and facilities	Admin functions worked successfully	Passed
Real-time updates	Availability should update after actions	Updates appeared successfully	Passed

The functional evaluation showed that the major system features worked as expected.

5.4.2 Usability Evaluation

Usability evaluation focused on how easy the system was to use. The system was evaluated based on navigation, interface clarity, mobile responsiveness, role-based dashboards, and ease of completing major tasks.

The evaluation showed that the system interface was understandable and organized according to user roles. Drivers could easily search for parking facilities and reserve spaces. Attendants could access the scanner and check-in/check-out functions. Operators could manage facilities, zones, spaces, attendants, and prices from their dashboard. Administrators could access monitoring and management functions.

The system was also responsive, allowing users to access it on both mobile phones and desktop computers. This was important because drivers and attendants are more likely to use mobile devices, while operators and administrators may prefer larger screens.

Table 5.5: Usability Evaluation Results

Usability Area	Evaluation Result
Ease of navigation	Users could move between major sections easily
Interface clarity	Buttons, forms, and dashboard sections were clear
Mobile responsiveness	The system adjusted well to mobile screens
Role-based experience	Each user accessed only relevant functions
Task completion	Users could complete reservation, check-in, checkout, and payment tasks
Overall usability	The system was suitable for basic parking management operations

5.4.3 Performance Evaluation

Performance evaluation focused on how quickly the system responded to user actions. The main areas evaluated included page loading, API response, reservation processing, check-in/check-out processing, payment simulation, and real-time availability updates.

The evaluation showed that the system responded within an acceptable time during normal use. Searching for facilities, creating reservations, checking in, checking out, and generating receipts were completed successfully. Real-time updates were also reflected after system events such as reservation, check-in, and checkout.

Table 5.6: Performance Evaluation Results

Performance Area	Result
Page loading	Pages loaded successfully within acceptable time
Facility search	Search results were displayed successfully
Reservation processing	Reservations were created without noticeable delay
QR verification	Check-in and checkout verification worked efficiently
Payment simulation	Simulated payment was processed successfully
Real-time update response	Availability updates appeared after system actions

5.4.4 Security Evaluation

Security evaluation focused on the protection of user accounts, system routes, and role-based functions. The system used password hashing, token-based authentication, role-based access control, and secure API middleware.

Passwords were not stored in plain text. JWT authentication was used to protect system routes after login. Role-based access control ensured that drivers, attendants, operators, and administrators could only access functions assigned to them.

For example, drivers could not access operator facility management functions, attendants could not approve facilities, and operators could not access administrator-only controls.

The Parka documentation identifies bcrypt password hashing, JWT protection, Helmet, CORS, and rate limiting as part of the security measures implemented in the system.

Table 5.7: Security Evaluation Results

Security Area	Evaluation Result

Security Area	Evaluation Result
Password protection	Passwords were hashed before storage
Authentication	JWT authentication protected user sessions
Role-based access	Users accessed only functions assigned to their roles
Protected routes	Restricted pages required valid login
API security	Security middleware was used to reduce common risks
System control	Administrator functions supported platform monitoring

5.4.5 Reliability and Data Integrity Evaluation

Reliability and data integrity evaluation focused on whether the system could maintain accurate records and prevent inconsistent parking operations. This was important because a parking management system must avoid double-booking, incorrect space status, and inaccurate session records.

The system used database relationships, constraints, status fields, and transaction-safe operations to maintain consistency. When a space was reserved, its status changed. When a vehicle checked in, the space became occupied. When the vehicle checked out and payment was completed, the space became available again.

The Parka documentation explains that transactional queries were used to prevent double-booking and race conditions during critical operations such as reservation and check-in.

Table 5.8: Reliability and Data Integrity Evaluation Results

Evaluation Area	Result
Reservation consistency	Reserved spaces were updated correctly
Session tracking	Check-in and checkout sessions were recorded
Space status updates	Spaces changed between available, reserved, and occupied states
Payment record accuracy	Successful payments were recorded

Evaluation Area	Result
Receipt generation	Receipts were created after payment
Double-booking prevention	Transaction-safe logic helped prevent inconsistent reservations

5.5 Evaluation Against Study Objectives

The system was evaluated against the study objectives to determine whether the project achieved its intended purpose.

5.5.1 Objective One: To Review the Current System and Identify Gaps

The first objective was achieved by analyzing the existing parking management practices in Kampala. The study identified that the current system was largely manual and lacked real-time availability information, advance reservation, digital payment tracking, centralized operator dashboards, and reliable records.

These findings confirmed the need for a digital parking management and reservation system.

5.5.2 Objective Two: To Design a System That Addresses the Identified Gaps

The second objective was achieved through the design of Parka. The system design included user roles, system architecture, database schema, use case design, context diagram, reservation workflow, check-in/check-out workflow, and payment workflow.

The design addressed the identified gaps by introducing real-time availability, parking reservation, QR verification, digital records, role-based dashboards, and operator facility management.

5.5.3 Objective Three: To Implement the Designed System

The third objective was achieved through the development of Parka as a working web-based system. The frontend was implemented using React, while the backend was implemented using Node.js and Express. PostgreSQL was used for data storage, and Socket.io was used for real-time communication.

The implemented system included driver, attendant, operator, and administrator portals. It also included reservation, QR check-in/check-out, payment simulation, receipt generation, notifications, and facility management.

5.5.4 Objective Four: To Evaluate the Effectiveness of the System

The fourth objective was achieved by evaluating the system in terms of functionality, usability, performance, security, reliability, and data integrity. The evaluation showed that the system performed its core functions successfully.

The system supported digital parking search, reservation, check-in, checkout, payment simulation, receipt generation, and real-time updates. This demonstrated that Parka addressed the major limitations identified in the existing parking management system.

Table 5.9: Evaluation of the System Against Study Objectives

Study Objective	How It Was Achieved
Review the current system and identify gaps	Existing manual parking practices were analyzed and gaps were identified
Design a system to address the gaps	Parka was designed with architecture, workflows, user roles, and database models
Implement the designed system	The system was developed using React, Node.js, Express, PostgreSQL, and Socket.io
Evaluate the effectiveness of the system	The system was tested and evaluated against functionality, usability, performance, security, and reliability

5.6 Discussion of Results

The results of the study showed that Parka addressed the major challenges identified in the existing parking management system. The discussion is presented under parking management findings, system functionality, system benefits, and system limitations.

5.6.1 Discussion of Parking Management Findings

The findings showed that the existing parking system in Kampala was mainly manual and lacked real-time information. Drivers experienced uncertainty when searching for parking spaces, while attendants and operators relied on manual processes to manage parking operations.

These findings confirmed the problem statement of the study. The lack of real-time availability and advance reservation contributed to time wastage, fuel consumption, congestion, and poor space utilization.

Parka responded to these issues by providing a digital platform where parking spaces could be searched, reserved, managed, and monitored through role-based interfaces.

5.6.2 Discussion of System Functionality

The system functionality showed that Parka successfully supported the main parking management operations. Drivers could search and reserve parking spaces, attendants could verify reservations and manage entry or exit, operators could manage facilities and spaces, and administrators could monitor the platform.

The reservation function directly addressed the absence of advance booking in the existing system. The QR-based check-in and checkout function improved verification and session tracking. The payment simulator and receipt generation improved transaction records and accountability.

The real-time update feature also improved the accuracy of parking availability information, allowing users to receive updated information after reservations, check-ins, and checkouts.

5.6.3 Discussion of System Benefits

The developed system provided several benefits to different users.

For drivers, Parka reduced uncertainty by allowing them to search for parking and reserve spaces before arrival. This improved convenience and reduced the need to move around physically while searching for parking.

For attendants, the system improved parking session handling through QR-based verification and digital check-in/check-out. This reduced manual errors and improved session tracking.

For operators, the system provided better visibility into parking facility operations. Operators could manage facilities, zones, spaces, attendants, prices, reservations, sessions, and payments from a digital dashboard.

For administrators, the system provided platform-level control through user management, operator verification, facility approval, and monitoring.

Overall, Parka provided a foundation for more organized and accountable parking management in Kampala.

5.6.4 Discussion of System Limitations

Although the system achieved its main objectives, some limitations were identified.

First, the system did not use physical parking sensors. Parking availability depended on digital updates from reservations, check-ins, and checkouts rather than automatic detection from hardware sensors.

Second, payment was simulated rather than integrated with live MTN Mobile Money or Airtel Money APIs. This was because live payment integration was outside the scope of the academic project.

Third, testing was done in a controlled academic environment rather than across a large number of real parking facilities in Kampala.

Fourth, the system required internet connectivity to function properly. In areas with poor connectivity, real-time updates may be affected.

Lastly, the system focused on core parking management features and did not include advanced features such as automatic number plate recognition, IoT sensors, artificial intelligence-based parking prediction, or full integration with city transport systems.

5.7 Challenges Encountered During Evaluation

Several challenges were encountered during system evaluation.

One challenge was the limited access to live parking facility data. Since many parking facilities do not maintain digital records, some system data had to be simulated for demonstration and testing purposes.

Another challenge was the absence of physical parking sensors. This required the system to depend on digital status updates from user actions such as reservation, check-in, and checkout.

Payment integration was also a challenge because live mobile money APIs require additional approval, configuration, and production access. As a result, a payment simulator was used.

Testing real-time updates also required multiple user roles to interact with the system at the same time. For example, a driver could reserve a space while an attendant checked in the vehicle and an operator monitored occupancy.

Despite these challenges, the system was successfully evaluated and demonstrated the ability to support real-time parking management and space reservation.

5.8 Summary of the Study

The study focused on the design and implementation of a real-time parking management and space reservation system for Kampala. The main problem addressed was the inefficiency of the existing manual parking management system, where drivers physically search for parking spaces without prior information about availability.

The study began by examining the background of parking management challenges in Kampala. It identified that the growing number of vehicles, limited organized parking infrastructure, and dependence on manual parking processes had contributed to time wastage, fuel consumption, traffic congestion, and poor user experience.

The literature review showed that traditional parking systems are simple but inefficient because they lack real-time information, reservation support, digital records, and centralized monitoring. The review also showed that smart parking systems can improve parking management through real-time availability, reservation, digital payment, and operator dashboards. However, many existing smart parking systems depend on expensive infrastructure such as sensors, automated gates, and license plate recognition systems, which may not be affordable for many parking operators in developing cities.

The methodology chapter explained the research design, population, sampling techniques, data collection methods, data analysis techniques, system development methodology, tools, technologies, testing methods, and ethical considerations. The study used a case study design supported by descriptive and developmental approaches. Agile methodology was used during system development because it allowed iterative development, testing, and improvement.

The system analysis, design, and implementation chapter presented the existing system, strengths and weaknesses of the manual parking process, functional requirements, non-functional requirements, user requirements, system users, architecture, design models, database design, implementation details, security, and deployment. Parka was implemented using React for the frontend, Node.js and Express for the backend, PostgreSQL for data storage, and Socket.io for real-time updates.

This chapter presented the research findings, system results, evaluation, and discussion of the developed system. The evaluation showed that Parka successfully supported parking search, availability display, reservation, QR-based check-in and check-out, payment simulation, receipt generation, notifications, facility management, and administrator monitoring.

5.9 Achievement of the Study Objectives

The study was guided by four specific objectives. The extent to which each objective was achieved is presented below.

5.9.1 Objective One: To Review the Current Parking Management System and Identify Gaps

This objective was achieved by analyzing the existing parking management practices in Kampala. The study found that the current parking system was largely manual and depended on physical searching, parking attendants, cash payments, and informal record keeping.

The major gaps identified included lack of real-time parking availability information, absence of advance reservation, poor session tracking, limited digital payment support, lack of centralized operator dashboards, and weak reporting. These findings justified the need for a digital parking management and space reservation system.

5.9.2 Objective Two: To Design a Real-Time Parking Management and Space Reservation System

This objective was achieved through the design of Parka. The system design included the system architecture, user roles, functional requirements, non-functional requirements, user requirements, database design, use case design, context diagram, reservation workflow, and check-in/check-out workflow.

The system was designed to support four major user roles: driver, parking attendant, parking facility operator, and system administrator. Each role was assigned specific functions to ensure proper separation of responsibilities and secure access control.

5.9.3 Objective Three: To Implement the Designed System Using Suitable Web Technologies

This objective was achieved by developing Parka as a working web-based application. The frontend was implemented using React, Vite, TypeScript, and Tailwind CSS. The backend was implemented using Node.js and Express, while PostgreSQL was used as the database. Socket.io was used to support real-time updates.

The implemented system included user authentication, role-based dashboards, parking facility search, vehicle management, reservation engine, QR-based check-in and check-out, fee calculation, payment simulation, receipt generation, notifications, facility management, attendant management, and administrator monitoring.

5.9.4 Objective Four: To Evaluate the Effectiveness of the Developed System

This objective was achieved through system evaluation. The system was evaluated in terms of functionality, usability, performance, security, reliability, and data integrity.

The evaluation showed that the system performed its core functions successfully. Users could register, log in, search for parking, reserve spaces, check in, check out, complete simulated payments, generate receipts, and receive updates. Operators could manage facilities, spaces, attendants, and pricing, while administrators could manage users and monitor the platform.

5.10 Conclusion

The study successfully designed and implemented Parka, a real-time parking management and space reservation system for Kampala. The system addressed the major limitations of the existing manual parking management system by providing digital parking search, real-time availability display, advance reservation, QR-based check-in and check-out, payment simulation, receipt generation, role-based dashboards, notifications, and administrative monitoring.

The developed system demonstrated that web-based technologies can be used to improve parking management in urban areas such as Kampala. By allowing drivers to view parking availability and reserve spaces before arrival, Parka reduces uncertainty and improves convenience. By enabling attendants to verify reservations and manage parking sessions digitally, the system improves operational accuracy. By providing operators with dashboards for managing facilities, spaces, attendants, and payments, the system improves visibility and accountability.

The study also showed that a cost-effective smart parking solution can be developed without depending heavily on expensive physical infrastructure such as sensors, automated gates, or license plate recognition cameras. Although these technologies can improve future versions of the system, Parka demonstrated that real-time parking management can begin with digital reservation, check-in, check-out, and database-driven availability updates.

Overall, the study achieved its main objective of designing and implementing a real-time parking management and space reservation system to improve parking efficiency in Kampala.

5.11 Recommendations

Based on the findings, system implementation, and evaluation, the following recommendations were made.

5.11.1 Adoption of Digital Parking Management Systems

Parking operators in Kampala should adopt digital parking management systems to improve parking availability tracking, reservation, session monitoring, payment accountability, and reporting. Manual methods are no longer sufficient for busy urban areas where drivers need timely parking information.

5.11.2 Integration of Real-Time Parking Availability

Parking facilities should provide real-time information about available, reserved, and occupied spaces. This would help drivers reduce the time spent searching for parking and improve overall traffic flow.

5.11.3 Introduction of Advance Parking Reservation

Parking operators should allow drivers to reserve spaces before arrival. Advance reservation improves convenience, reduces uncertainty, and enables operators to plan space usage more effectively.

5.11.4 Digital Payment Integration

Future parking systems should integrate live payment options such as MTN Mobile Money, Airtel Money, debit cards, or other digital payment channels. This would improve transparency, reduce cash handling, and support accurate revenue records.

5.11.5 Training of Parking Attendants and Operators

Parking attendants and operators should be trained on how to use digital parking systems. Training would improve system adoption and reduce resistance to change from manual to digital methods.

5.11.6 Support from Urban Authorities

Urban authorities such as Kampala Capital City Authority should support smart parking initiatives as part of urban mobility improvement. Digital parking data can help in planning, monitoring, enforcement, and decision-making.

5.11.7 Continuous System Improvement

The system should be improved continuously based on user feedback, operational needs, and technological changes. This would ensure that the system remains useful, secure, and scalable.

5.12 Areas for Future Work

Although Parka achieved its main objectives, several areas can be improved in future versions of the system.

5.12.1 Integration with Live Mobile Money APIs

Future versions should integrate live MTN Mobile Money and Airtel Money APIs to allow real parking payments instead of simulated transactions. This would make the system more practical for commercial deployment.

5.12.2 Integration of IoT Parking Sensors

The system can be improved by integrating IoT sensors to automatically detect whether a parking space is occupied or available. This would improve the accuracy of real-time availability updates.

5.12.3 License Plate Recognition

Future development can include automatic license plate recognition to support automated vehicle identification during entry and exit. This would reduce the need for manual QR scanning in advanced deployments.

5.12.4 Google Maps and Route Guidance

The system can be improved by integrating route guidance to help drivers navigate to selected parking facilities. This would improve user convenience and reduce time spent locating facilities.

5.12.5 Artificial Intelligence-Based Parking Prediction

Artificial intelligence can be used to predict parking demand based on time, location, day of the week, events, and historical parking patterns. This would help operators plan better and help drivers choose the best parking options.

5.12.6 Expansion Beyond Kampala

The system can be expanded to other urban areas in Uganda such as Entebbe, Wakiso, Jinja, Mbarara, and Mukono. This would allow the solution to support wider urban parking management.

5.12.7 Native Mobile Application

Although Parka was implemented as a web-based system, future versions can include native Android and iOS mobile applications to improve mobile user experience and support push notifications.

5.12.8 Advanced Analytics Dashboard

The operator and administrator dashboards can be improved with advanced analytics such as revenue trends, occupancy heat maps, peak-hour reports, customer behavior, and facility performance comparisons.

5.13 Final Remarks

Parka demonstrated that a web-based real-time parking management and space reservation system can help address the limitations of manual parking systems in Kampala. The system provided a practical approach to parking search, reservation, QR-based check-in and check-out, payment simulation, receipt generation, and facility management.

The project contributed to the application of information systems in solving real urban mobility challenges. It also provided a foundation for future smart parking solutions that can be enhanced with live payment integration, IoT sensors, license plate recognition, artificial intelligence, and wider deployment.

The successful implementation and evaluation of Parka showed that digital transformation can improve parking management, reduce uncertainty for drivers, support better operator control, and contribute to more organized urban transport systems.

REFERENCES

Adam Smith International. (n.d.). *Uganda: Improvement of traffic congestion-flow in Kampala City*. United Nations Economic and Social Commission for Asia and the Pacific. https://sdghelpdesk.unescap.org/sites/default/files/2018-03/Uganda%20Improvement%20of%20Traffic%20Congestion-Flow%20in%20Kampala%20City.pdf

Barone, R. E., Giuffrè, T., Siniscalchi, S. M., Morgano, M. A., & Tesoriere, G. (2014). Architecture for parking management in smart cities. *IET Intelligent Transport Systems, 8*(5), 445–452. https://doi.org/10.1049/iet-its.2013.0045

Biyik, C., Allam, Z., Pieri, G., Moroni, D., O'Fraifer, M., O'Connell, E., Olariu, S., & Khalid, M. (2021). Smart parking systems: Reviewing the literature, architecture and ways forward. *Smart Cities, 4*(2), 623–642. https://doi.org/10.3390/smartcities4020032

Geng, Y., & Cassandras, C. G. (2013). New "smart parking" system based on resource allocation and reservations. *IEEE Transactions on Intelligent Transportation Systems, 14*(3), 1129–1139. https://doi.org/10.1109/TITS.2013.2252428

Litman, T. (2025). *Parking management: Comprehensive implementation guide*. Victoria Transport Policy Institute. https://www.vtpi.org/park_man_comp.pdf

Mugizi, B., Murindanyi, S., Nakacwa, O., & Katumba, A. (2026). *Intelligent traffic surveillance for real-time vehicle detection, license plate recognition, and speed estimation*. arXiv. https://arxiv.org/abs/2601.00344

Multiplex Limited. (n.d.-a). *Multiplex street parking*. https://multiplexstreetparking.com/

Multiplex Limited. (n.d.-c). *Plex Parking*. Google Play. https://play.google.com/store/apps/details?id=com.multiplex.parking

Shoup, D. C. (2006). Cruising for parking. *Transport Policy, 13*(6), 479–486. https://doi.org/10.1016/j.tranpol.2006.05.005

APPENDIX A: RESEARCH QUESTIONNAIRE

RESEARCH QUESTIONNAIRE FOR DRIVERS AND PARKING USERS

Topic: Design and Implementation of a Real-Time Parking Management and Space Reservation System: A Case Study of Kampala

Dear Respondent,

I am a student of **Nkumba University**, pursuing a Bachelor's Degree in Information Systems and Technology. I am conducting a study on the **design and implementation of a real-time parking management and space reservation system**, with Kampala as the case study.

The purpose of this questionnaire is to collect information about the current parking management challenges, user experiences, and the need for a digital parking management and space reservation system. The information provided will be used strictly for academic purposes and will be treated with confidentiality.

Participation in this study is voluntary. You are kindly requested to answer the questions honestly by ticking the most appropriate option or writing your response where required.

SECTION A: DEMOGRAPHIC INFORMATION

1. Gender

☐ Male

☐ Female

☐ Prefer not to say

2. Age group

☐ Below 20 years

☐ 20–29 years

☐ 30–39 years

☐ 40–49 years

☐ 50 years and above

3. Occupation

☐ Student

☐ Business person

☐ Employee

☐ Driver / Transport operator

☐ Self-employed

☐ Other, specify: ________________________________

4. Do you own or regularly use a vehicle?

☐ Yes

☐ No

5. Type of vehicle commonly used

☐ Private car

☐ Taxi

☐ Motorcycle

☐ Delivery vehicle

☐ Company vehicle

☐ Other, specify: ________________________________

SECTION B: CURRENT PARKING PRACTICES

6. How often do you look for parking space in Kampala?

☐ Daily

☐ Several times a week

☐ Once a week

☐ Occasionally

☐ Rarely

7. Where do you usually park when in Kampala?

☐ Street parking

☐ Private parking yard

☐ Shopping mall parking

☐ Office parking

☐ Public parking facility

☐ Other, specify: ________________________________

8. How do you usually find parking space?

☐ By physically moving around and searching

☐ By asking parking attendants

☐ Through previous knowledge of parking areas

☐ Through phone calls

☐ Through a mobile/web application

☐ Other, specify: ________________________________

9. Do you usually know whether parking space is available before reaching the parking location?

☐ Yes

☐ No

☐ Sometimes

10. How is parking payment usually made?

☐ Cash

☐ Mobile money

☐ Card payment

☐ Monthly parking sticker/subscription

☐ Other, specify: ________________________________

11. Do you usually receive a receipt or proof of payment after parking?

☐ Yes

☐ No

☐ Sometimes

SECTION C: PARKING CHALLENGES

12. Do you experience difficulty finding parking space in Kampala?

☐ Yes

☐ No

☐ Sometimes

13. How long do you usually spend searching for parking space?

☐ Less than 5 minutes

☐ 5–10 minutes

☐ 11–20 minutes

☐ 21–30 minutes

☐ More than 30 minutes

14. Which of the following parking challenges do you experience? Tick all that apply.

☐ Lack of available parking spaces

☐ Lack of real-time information on available parking

☐ High parking fees

☐ Poor parking organization

☐ Time wasted searching for parking

☐ Cash payment inconvenience

☐ Lack of receipts or payment records

☐ Insecurity of parked vehicles

☐ Congestion around parking areas

☐ Other, specify: ________________________________

15. How does searching for parking affect you? Tick all that apply.

☐ Wastes time

☐ Increases fuel consumption

☐ Causes stress or frustration

☐ Delays arrival at destination

☐ Contributes to traffic congestion

☐ Increases transport costs

☐ Other, specify: ________________________________

16. In your opinion, what is the biggest parking problem in Kampala?

SECTION D: DIGITAL PARKING SYSTEM NEEDS

17. Have you ever used a digital system or mobile application to find or pay for parking?

☐ Yes

☐ No

18. Would you use a digital system that shows available parking spaces in real time?

☐ Yes

☐ No

☐ Maybe

19. Would you use a system that allows you to reserve parking space before arrival?

☐ Yes

☐ No

☐ Maybe

20. Which features would you expect in a digital parking management system? Tick all that apply.

☐ User registration and login

☐ View nearby parking facilities

☐ View real-time parking availability

☐ Reserve parking space before arrival

☐ Receive reservation code or QR code

☐ Digital check-in and check-out

☐ Mobile money payment

☐ Receipt generation

☐ Parking fee calculation

☐ Notifications and reminders

☐ Directions to parking facility

☐ Vehicle registration

☐ Customer support

☐ Other, specify: ________________________________

21. How useful would a real-time parking availability system be to you?

☐ Very useful

☐ Useful

☐ Not sure

☐ Not useful

22. How useful would an advance parking reservation feature be to you?

☐ Very useful

☐ Useful

☐ Not sure

☐ Not useful

23. Would you prefer paying parking fees digitally instead of cash?

☐ Yes

☐ No

☐ Maybe

24. Which digital payment method would you prefer?

☐ MTN Mobile Money

☐ Airtel Money

☐ Debit/Credit card

☐ Bank transfer

☐ Other, specify: ________________________________

SECTION E: SYSTEM EVALUATION AND USER ACCEPTANCE

25. If a system like Parka was introduced, how likely would you be to use it?

☐ Very likely

☐ Likely

☐ Not sure

☐ Unlikely

26. What device would you most likely use to access the system?

☐ Smartphone

☐ Tablet

☐ Laptop/Desktop computer

☐ Other, specify: ________________________________

27. What would encourage you to use a digital parking reservation system? Tick all that apply.

☐ Easy to use

☐ Accurate parking availability

☐ Affordable parking fees

☐ Secure payments

☐ Reliable reservation confirmation

☐ Mobile money support

☐ Clear receipts

☐ Fast check-in and check-out

☐ Good customer support

☐ Other, specify: ________________________________

28. What concerns would you have about using a digital parking system? Tick all that apply.

☐ Internet connectivity problems

☐ Payment security

☐ Incorrect availability information

☐ Difficulty using the system

☐ Privacy of personal information

☐ Extra service charges

☐ System failure during check-in/check-out

☐ Other, specify: ________________________________

29. In your opinion, can a real-time parking management and reservation system improve parking efficiency in Kampala?

☐ Yes

☐ No

☐ Not sure

30. Explain your answer to Question 29.

____

____

____

SECTION F: GENERAL COMMENTS

31. What improvements would you recommend for parking management in Kampala?

____

____

____

32. Any additional comment about digital parking systems?

Thank you for participating in this study.

APPENDIX B: INTERVIEW GUIDE

INTERVIEW GUIDE FOR PARKING ATTENDANTS AND PARKING FACILITY OPERATORS

Topic: Design and Implementation of a Real-Time Parking Management and Space Reservation System: A Case Study of Kampala

Dear Respondent,

I am a student of **Nkumba University**, pursuing a Bachelor's Degree in Information Systems and Technology. I am conducting a study on the **design and implementation of a real-time parking management and space reservation system**, with Kampala as the case study.

The purpose of this interview guide is to collect information from parking attendants and parking facility operators about the current parking management practices, challenges faced in managing parking spaces, and the need for a digital parking management and space reservation system.

The information provided will be used strictly for academic purposes and will be treated with confidentiality. Participation in this interview is voluntary, and your responses will help in understanding how parking management can be improved through the proposed Parka system.

SECTION A: BACKGROUND INFORMATION

1. Respondent category

☐ Parking attendant

☐ Parking facility operator

☐ Parking manager/supervisor

☐ Other, specify: ________________________________

2. How long have you been involved in parking management or parking operations?

☐ Less than 1 year

☐ 1–3 years

☐ 4–6 years

☐ More than 6 years

3. What type of parking facility do you manage or work in?

☐ Street parking

☐ Private parking yard

☐ Shopping mall parking

☐ Office parking facility

☐ Public parking facility

☐ Other, specify: ________________________________

4. What is your main responsibility in parking operations?

SECTION B: CURRENT PARKING MANAGEMENT PRACTICES

5. How are parking spaces currently allocated to drivers?

6. How do you know whether parking spaces are available or occupied?

7. Do drivers usually know whether parking space is available before they arrive?

☐ Yes

☐ No

☐ Sometimes

8. How are vehicle entry and exit currently managed?

9. How are parking payments currently handled?

☐ Cash

☐ Mobile money

☐ Card payment

☐ Monthly subscription/sticker

☐ Other, specify: ________________________________

10. How are parking records currently kept?

☐ Paper records

☐ Manual receipts

☐ Computer system

☐ Mobile application

☐ Records are not properly kept

☐ Other, specify: ________________________________

SECTION C: CHALLENGES IN THE EXISTING SYSTEM

11. What challenges do you face when managing parking spaces?

12. What challenges do drivers usually report when looking for parking?

13. Do you experience difficulties in tracking occupied and available parking spaces?

☐ Yes

☐ No

☐ Sometimes

14. If yes, explain the difficulties experienced.

15. What challenges do you face when handling parking payments?

16. Are there cases of disputes with drivers over parking duration, fees, or receipts?

☐ Yes

☐ No

☐ Sometimes

17. If yes, what usually causes such disputes?

18. What challenges are experienced in monitoring attendants, parking sessions, or revenue?

SECTION D: NEED FOR A DIGITAL PARKING MANAGEMENT SYSTEM

19. Do you think a digital parking management system would improve parking operations?

☐ Yes

☐ No

☐ Not sure

20. Explain your answer.

21. Would real-time parking availability help drivers and parking operators?

☐ Yes

☐ No

☐ Not sure

22. Explain your answer.

23. Would advance parking reservation be useful in your parking facility or parking area?

☐ Yes

☐ No

☐ Not sure

24. Explain your answer.

---

25. Which features would be useful in a digital parking management system? Tick all that apply.

☐ Parking space availability tracking

☐ Parking reservation

☐ QR-based check-in and check-out

☐ Digital payment

☐ Receipt generation

☐ Attendant management

☐ Revenue tracking

☐ Facility dashboard

☐ Reports and analytics

☐ Notifications

☐ Customer support

☐ Other, specify: ________________________________

SECTION E: SYSTEM DESIGN AND FUNCTIONALITY EXPECTATIONS

26. What information should be shown to drivers before they arrive at a parking facility?

27. What information should be captured when a vehicle enters a parking facility?

28. What information should be captured when a vehicle exits a parking facility?

29. How should the system help attendants during check-in and check-out?

30. How should the system help operators manage parking facilities?

31. What kind of reports would be useful to parking operators or managers?

SECTION F: DIGITAL PAYMENT AND RECORD KEEPING

32. Would digital payment improve parking fee collection?

☐ Yes

☐ No

☐ Not sure

33. Explain your answer.

34. Which payment methods would be most suitable for parking users in Kampala?

☐ Cash

☐ MTN Mobile Money

☐ Airtel Money

☐ Card payment

☐ Bank transfer

☐ Monthly subscription/sticker

☐ Other, specify: ________________________________

35. How important is receipt generation in parking management?

☐ Very important

☐ Important

☐ Not sure

☐ Not important

36. Why is receipt generation important or not important?

SECTION G: SYSTEM ADOPTION AND CHALLENGES

37. Would parking attendants and operators be willing to use a digital parking system?

☐ Yes

☐ No

☐ Not sure

38. What may encourage attendants and operators to use such a system?

____

____

39. What challenges may affect the use of a digital parking system? Tick all that apply.

☐ Internet connectivity problems

☐ Lack of smartphones or computers

☐ Limited digital skills

☐ Resistance to change

☐ Cost of implementation

☐ Payment security concerns

☐ System failure or downtime

☐ Incorrect parking availability information

☐ Other, specify: ________________________________

40. What support or training would be needed for attendants and operators to use the system effectively?

____

____

____

SECTION H: GENERAL RECOMMENDATIONS

41. What improvements would you recommend for parking management in Kampala?

42. What advice would you give regarding the design of a real-time parking management and reservation system?

43. Any additional comments about parking management or digital parking systems?

Thank you for participating in this interview.

APPENDIX C: SCREENSHOTS OF THE SYSTEMS

Figure 9: Login screen on a desktop version

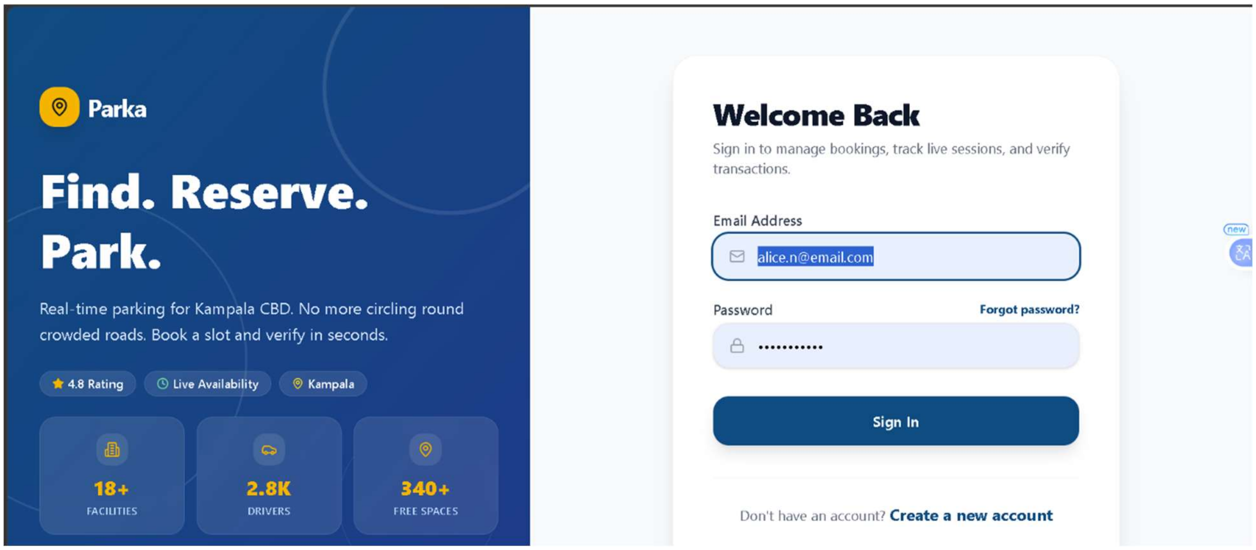


Figure 10:Attendant Dashboard

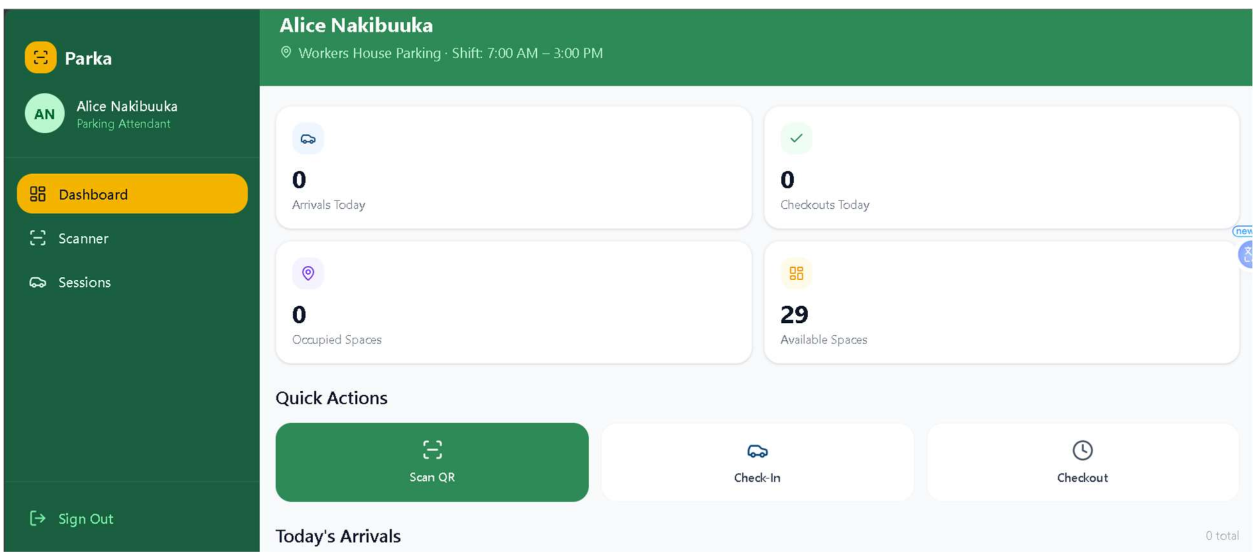


Figure 11: Driver Reservation comfirmed screen proving a QR code

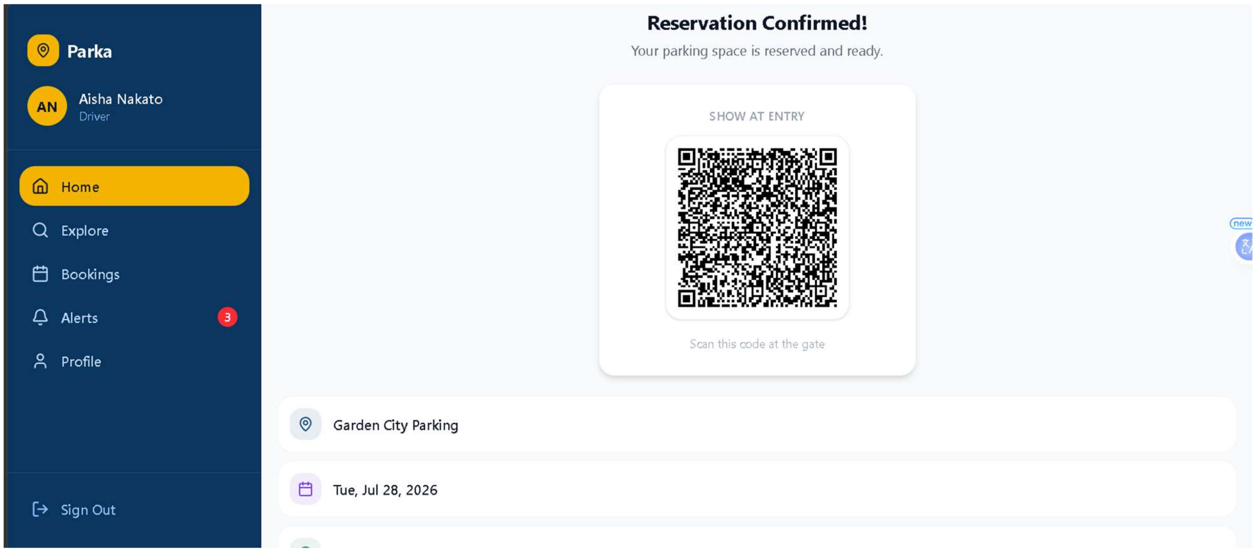


Figure 12:Driver notification screen

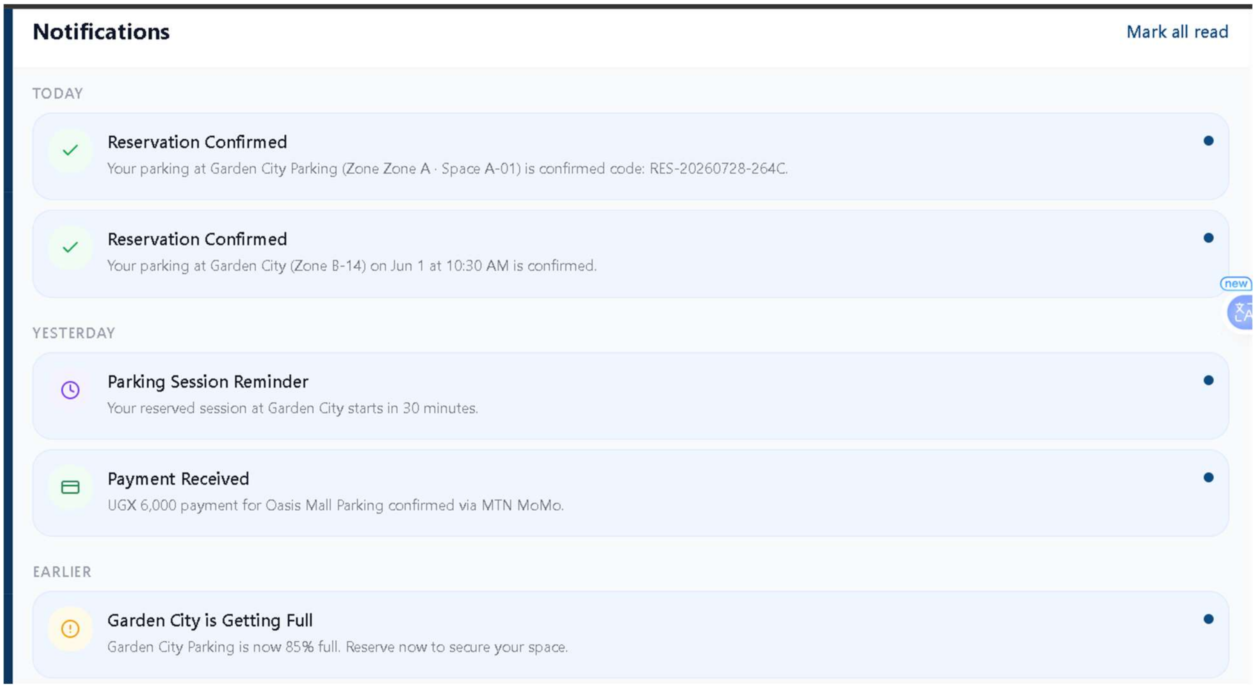


Figure 13: Driver Bookings

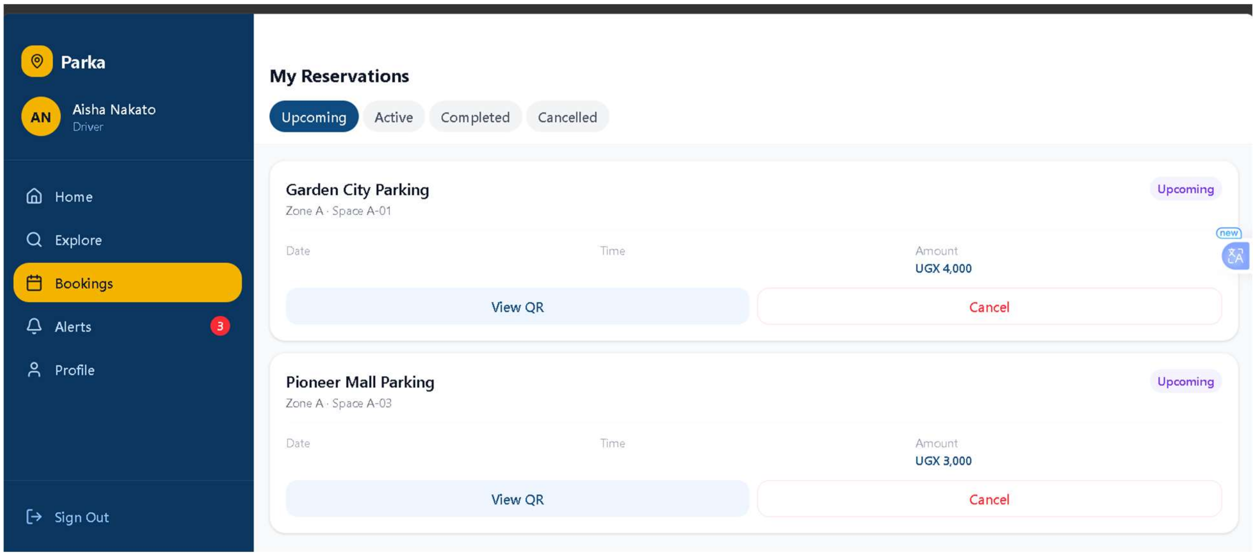


Figure 14:Driver dashboard

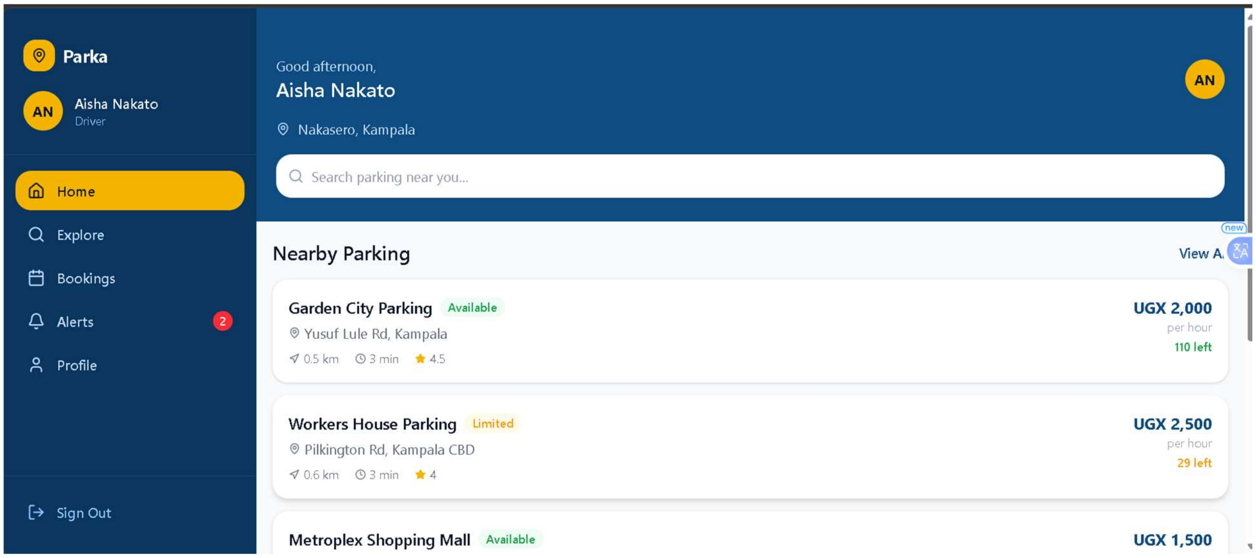


Figure 15Driver profile screen

Figure 16:Operator Facilities managements

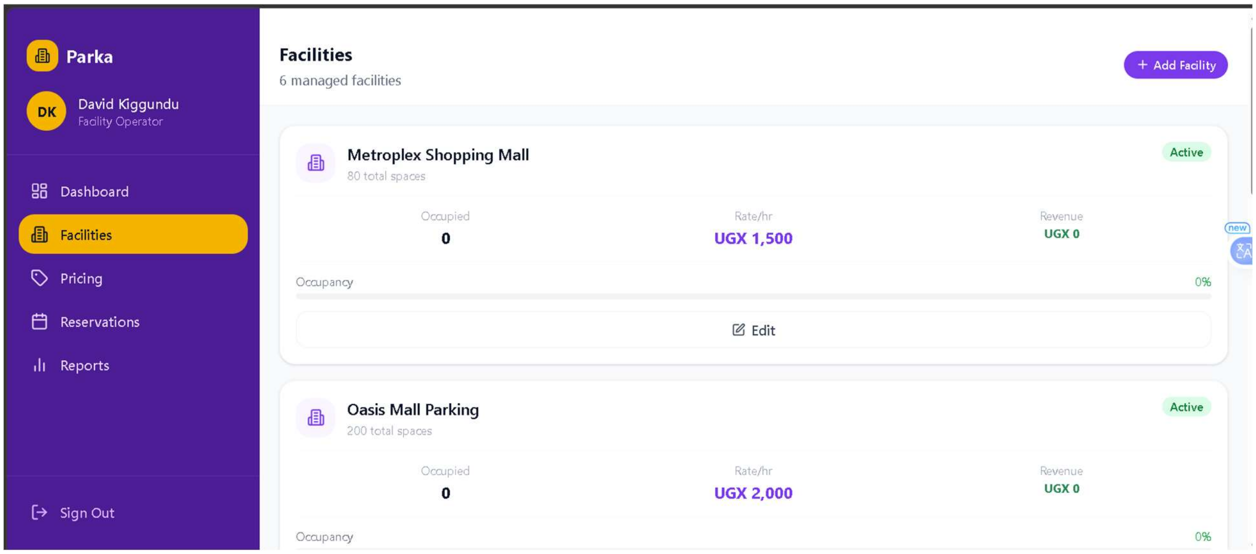


Figure 17:Operator Dashboard

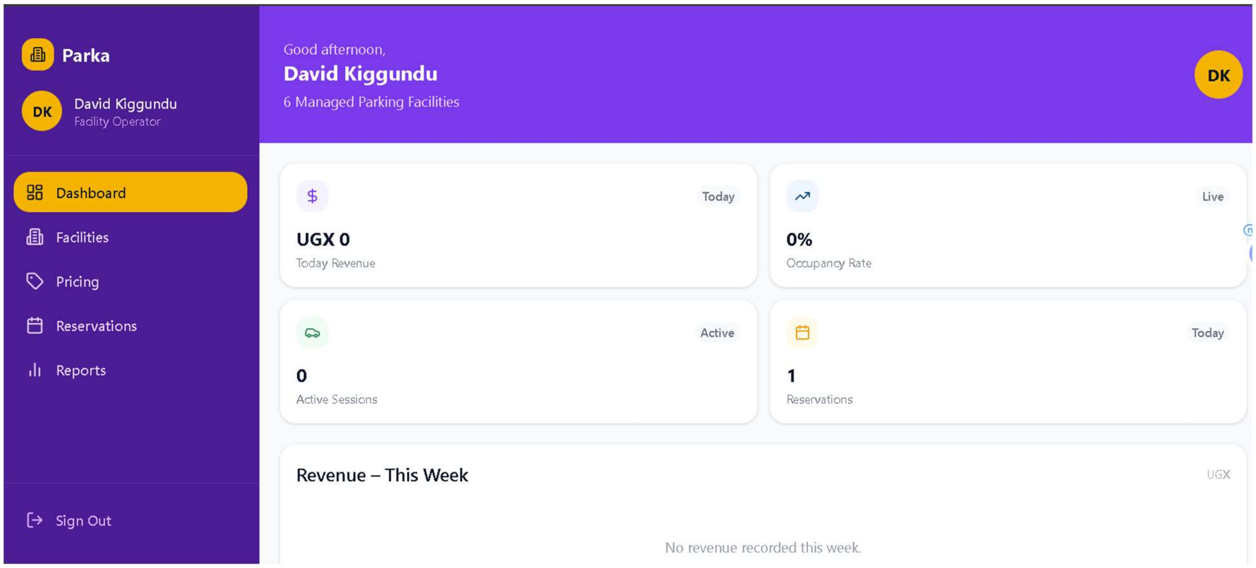


Figure 18:Operator Pricing management

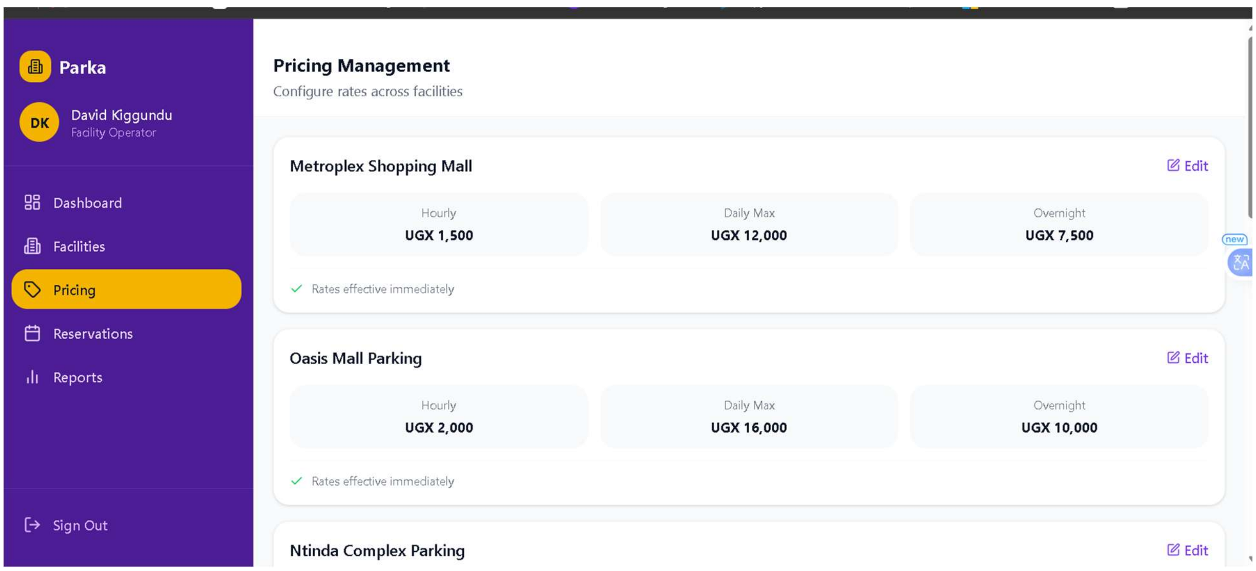


Figure 19:operator reservation screen

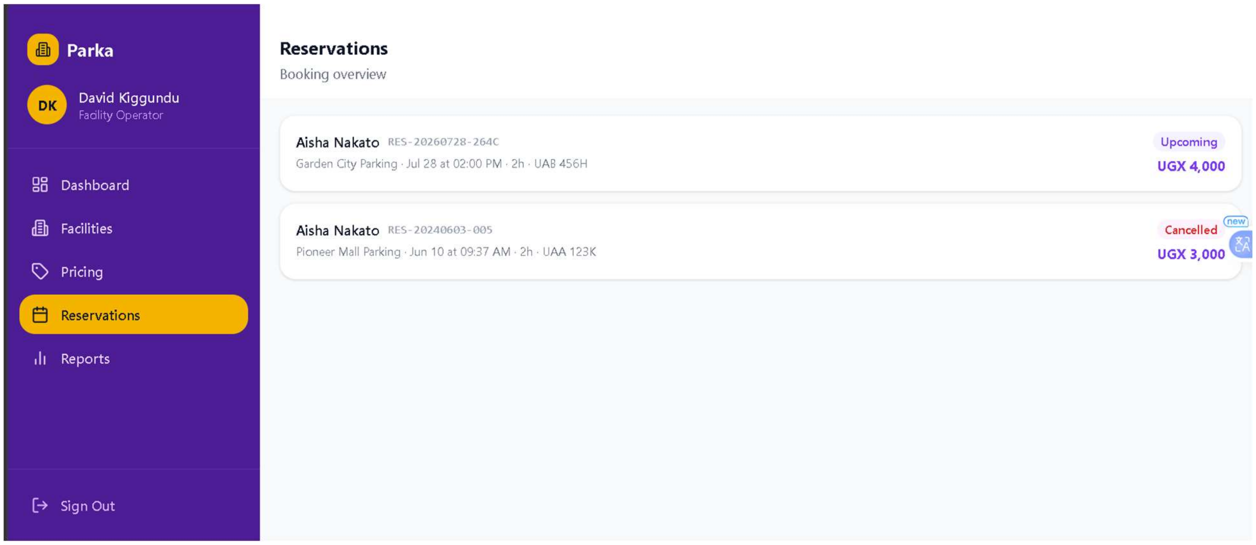


Figure 20:Super admin/Admin dashboard